

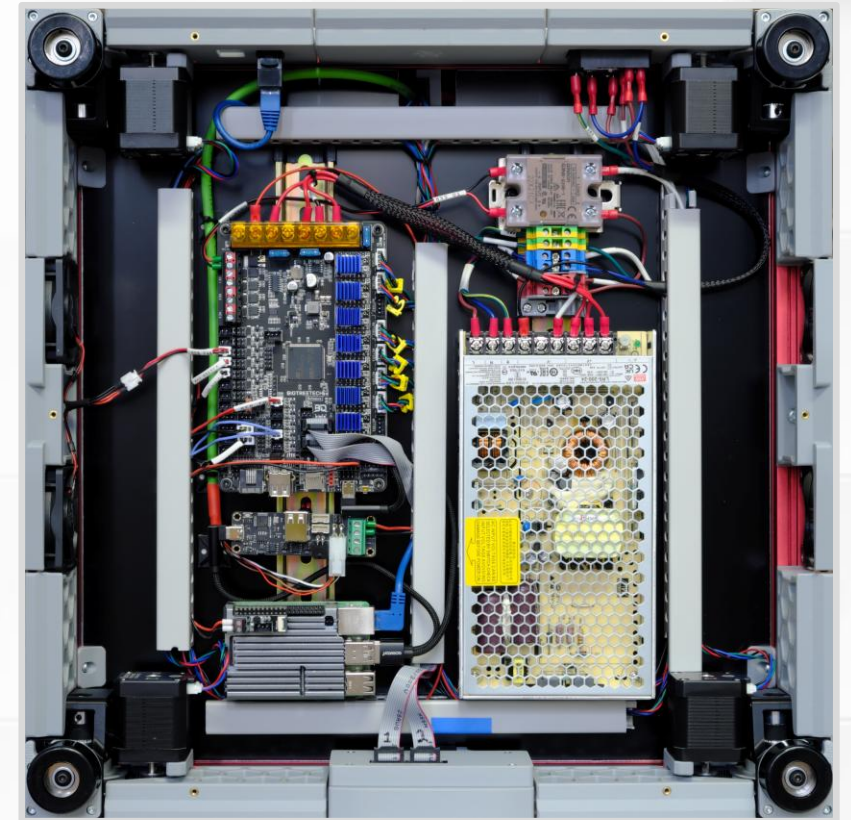
An Introduction to 3D Printing

Part 1: FFF Printers

*Process, Limitations, Dimensional Accuracy, and
Advantages*

About Me

- Hobby woodworking.
- Built my own printer to learn 3D printing.
- Voron 2.4 with over 6k hours of print time.



([Voron 2.4 GitHub](#))

Websites

- 3D designs.
printables.com/@FlyingGyroscope
- Get this guide.
github.com/FlyingGyroscope



Goals & Audience

- Training guide for students.
- Learn *strengths* and *weaknesses* of 3D printing.
- Make great prints.



Outline

Part 1: Printers

Process

Limitations

Dimensional Accuracy

Advantages

Part 2: Practical Application

Examples

Critical Thinking

Design

Part 3: Science

Vibrations and Waves

Linear Motion

Thermal Expansion

3D Printing is a Process

1) Start with a 3D design.

- Check the license for restrictions when you download.
- *STL*. Approximates shapes with triangles.
- *STEP*. Real geometry. Great for 3D modeling.
- *3MF*. Compressed container, like a zip file.
Multiple objects, slicer settings, multi-color settings, ...

2) Slice and process the design with software.

- Printers do not accept 3D models!
- “Slice” objects into vertically-stacked layers.
- Creates G-code file. List of toolpaths that encode coordinates, speeds, temperatures, ...

3D Printing is a Process

3) **Print.**

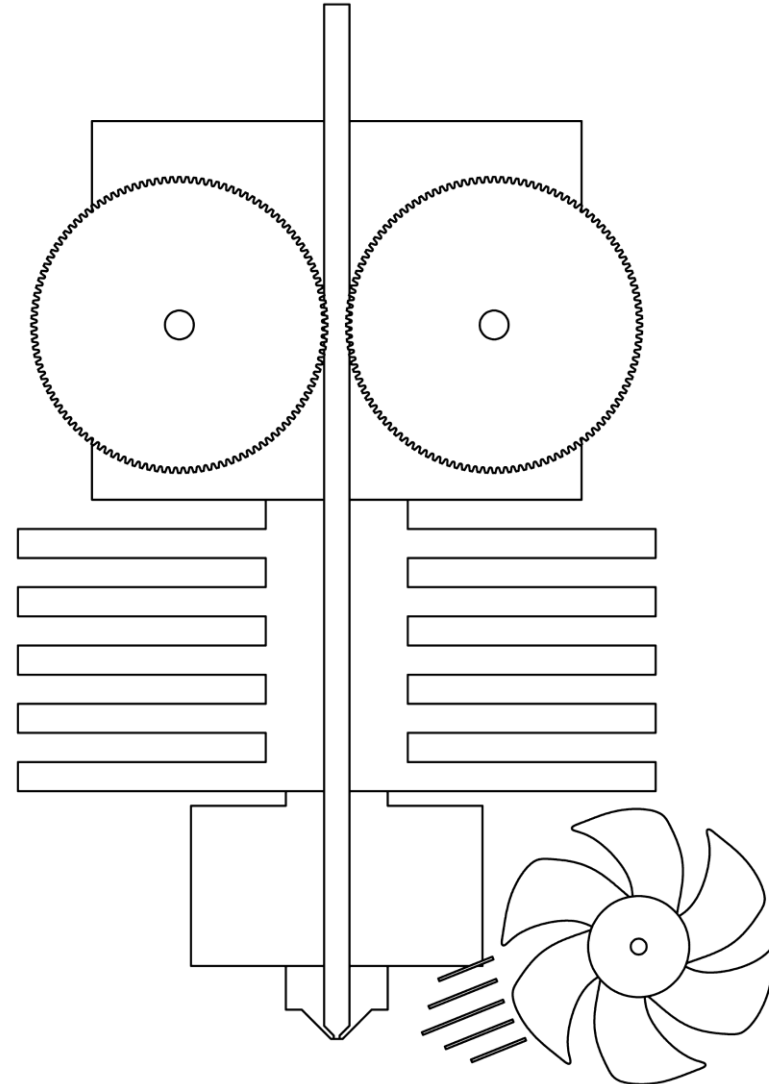
- Do you need to change the design?
- Change print settings?

4) **Post-process.**

- Remove supports.
- Assemble.
- Glue.
- Sand or file.
- Paint.

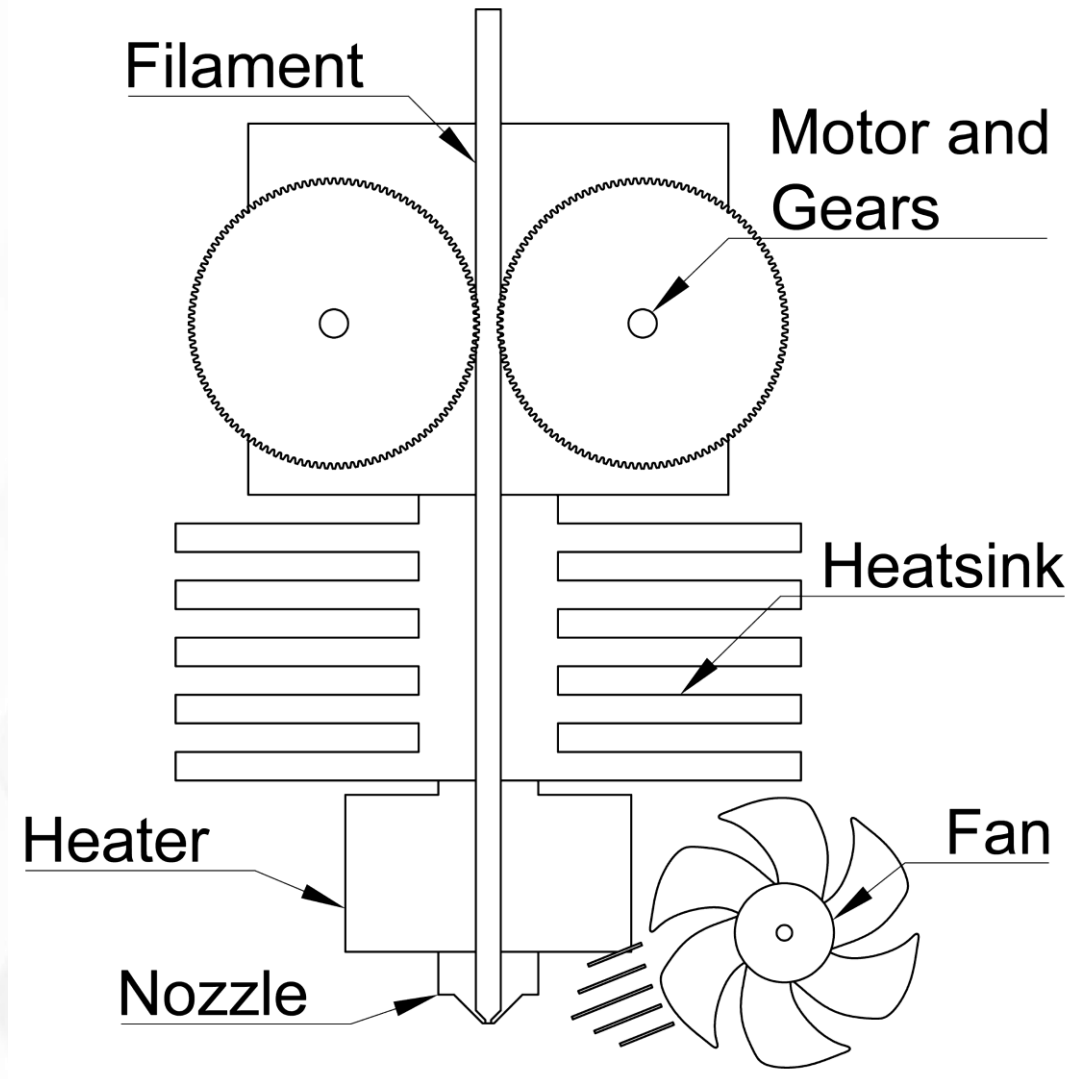
Pushing Plastic

- What kind of 3D printing?
- Filament and extrusion-based additive manufacturing.
- **FFF. Fused filament fabrication.**
- FDM™. Fused deposition modeling.
 - Fundamentally the same but owned by Stratasys.
- Toolhead. Moves around and deposits plastic.



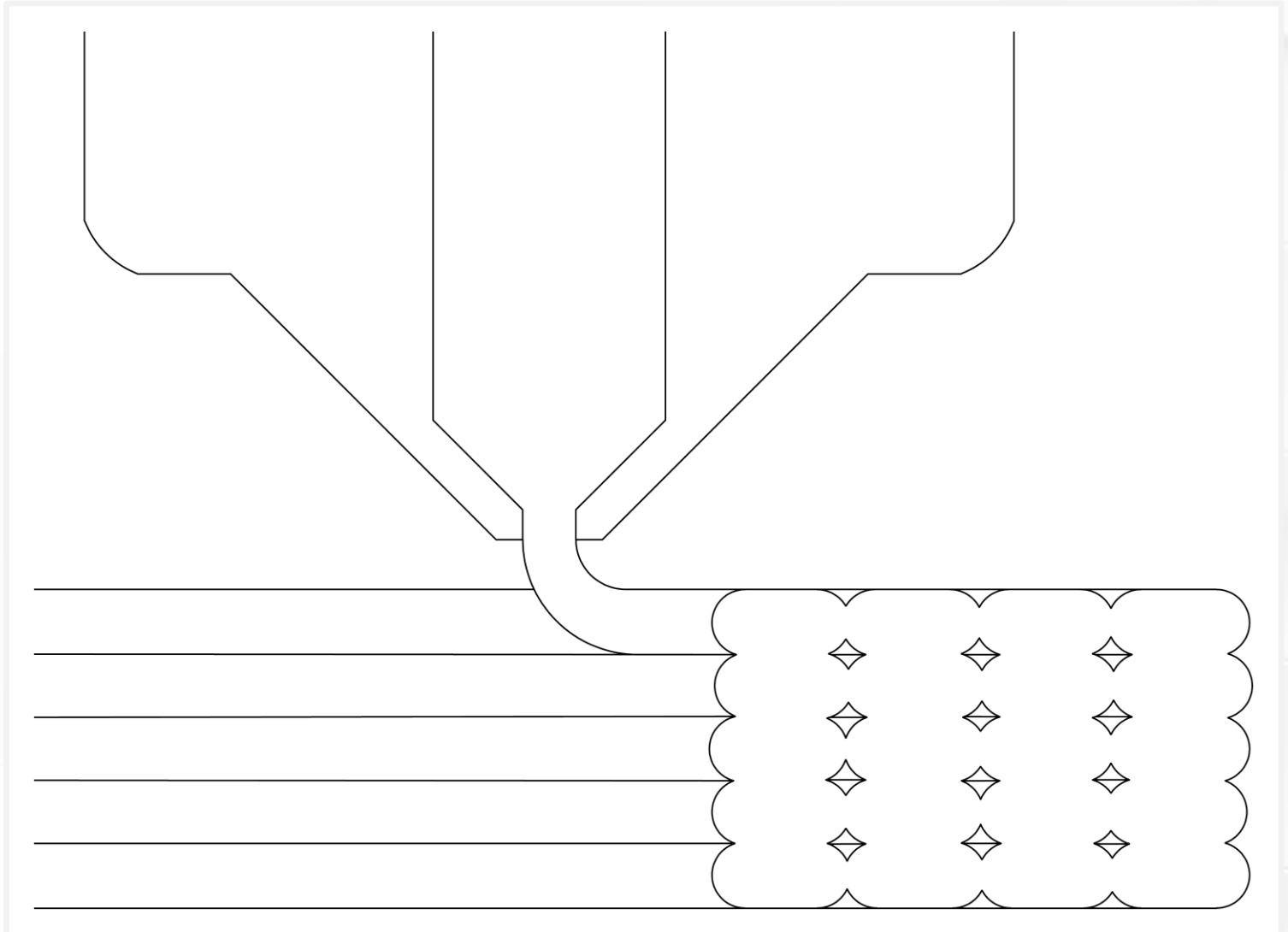
Pushing Plastic

- Filament. Long, continuous plastic rod.
- Motor and gears grab and push filament.
- Heatsink keeps heat in the hotend (heater + nozzle).
- Heater softens filament at the glass transition temperature.
- Nozzle opening shapes plastic.
- Fan cools hot plastic.



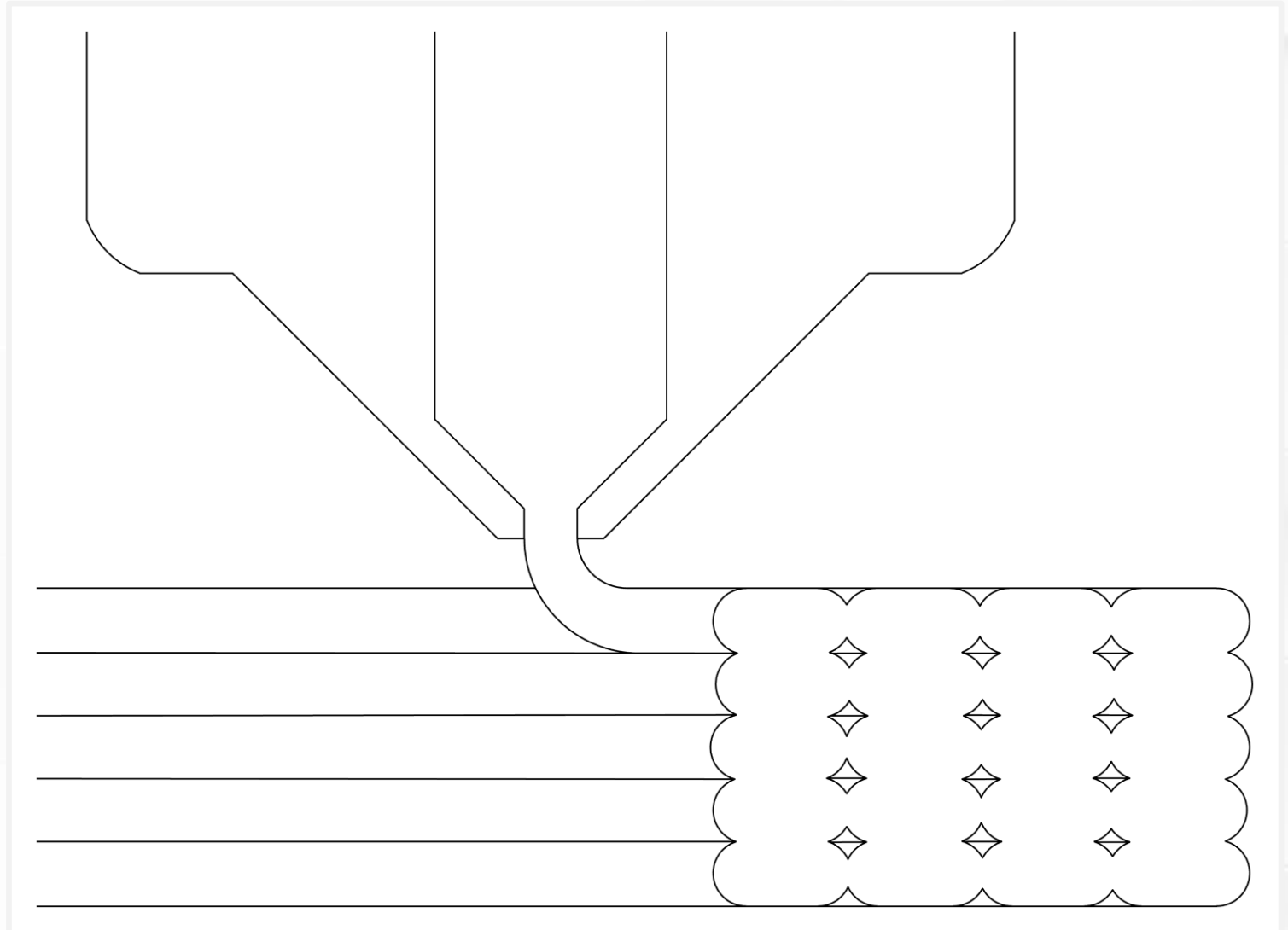
Extrusions

- Toolhead deposits lines called **extrusions**.
- Best to squish extrusions into place.
- Lock in place as they cool.



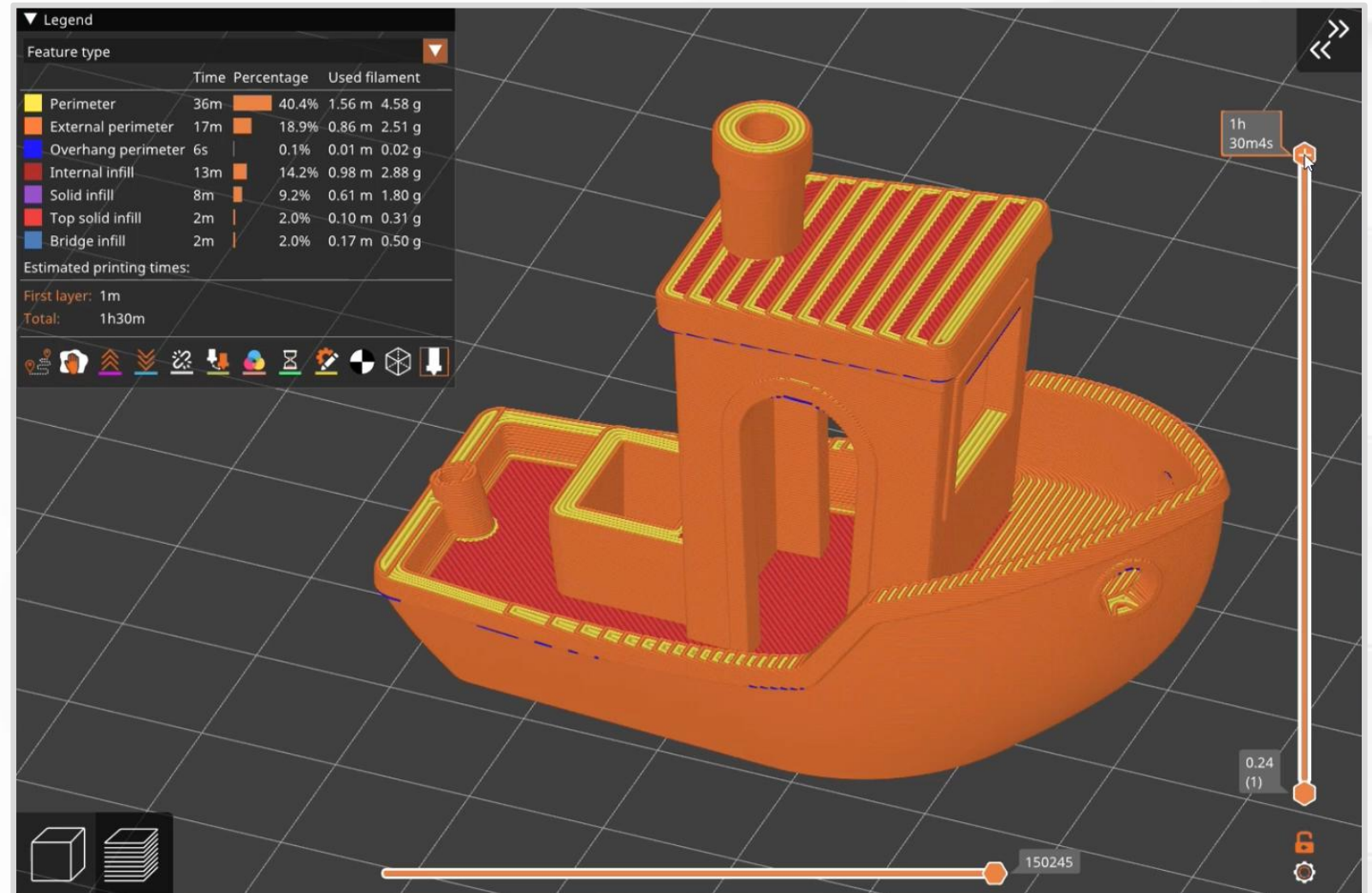
Stacking Layers

- Printers *simulate* 3D shapes by stacking 2D layers.
- You could call it 2.5D printing.



Slicer Layers

Let's see how PrusaSlicer handles layers.



(Animation,
FlyingGyroscope GitHub)

Slicer Layers

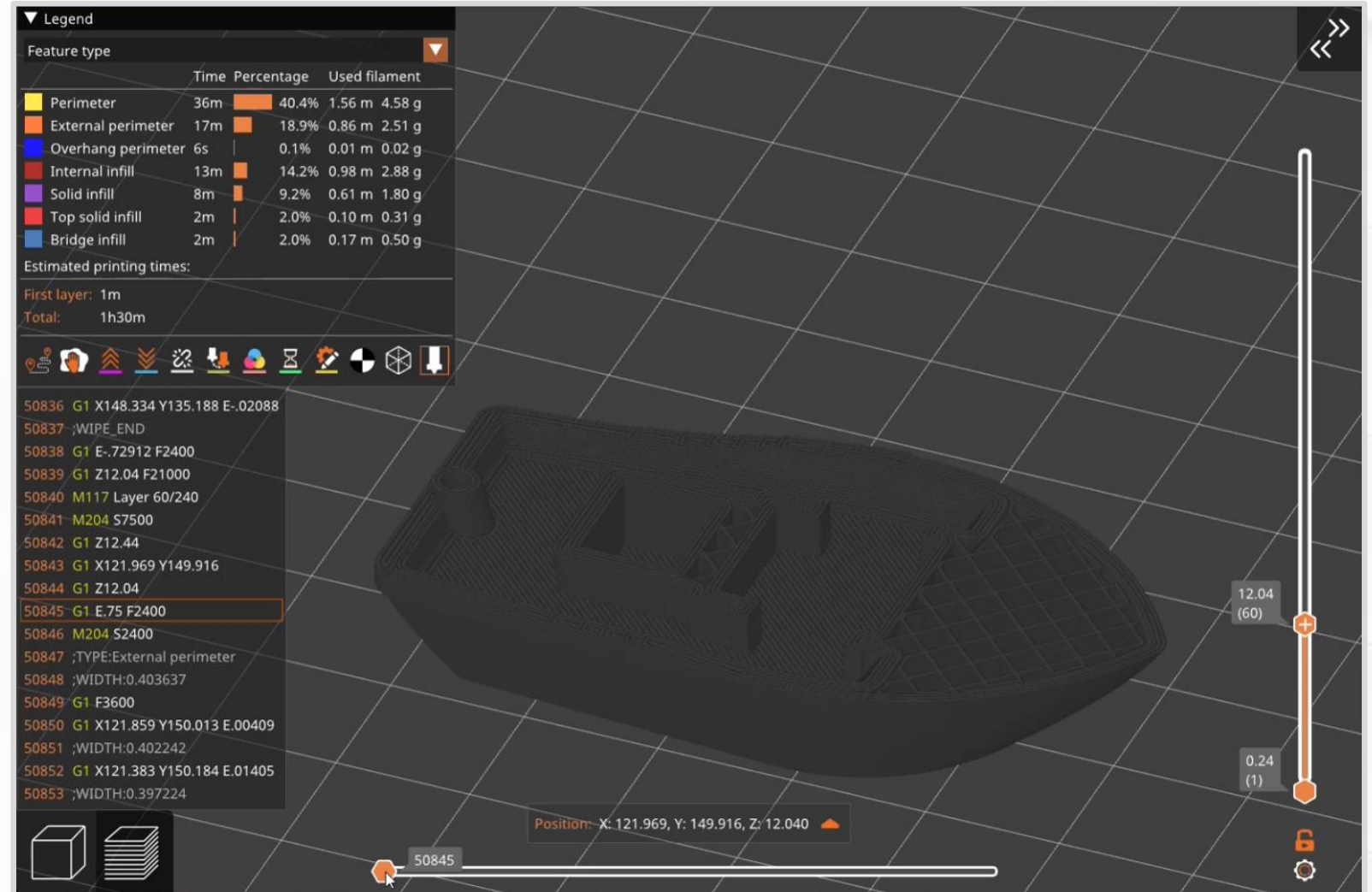
- Sort extrusions by shape and geometry.
- Color-coded.
- Adjust settings for each category.

▼ Legend					
Feature type ▼					
		Time	Percentage	Used filament	
	Perimeter	36m		40.4%	1.56 m 4.58 g
	External perimeter	17m		18.9%	0.86 m 2.51 g
	Overhang perimeter	6s		0.1%	0.01 m 0.02 g
	Internal infill	13m		14.2%	0.98 m 2.88 g
	Solid infill	8m		9.2%	0.61 m 1.80 g
	Top solid infill	2m		2.0%	0.10 m 0.31 g
	Bridge infill	2m		2.0%	0.17 m 0.50 g
Estimated printing times:					
First layer: 1m					
Total: 1h30m					

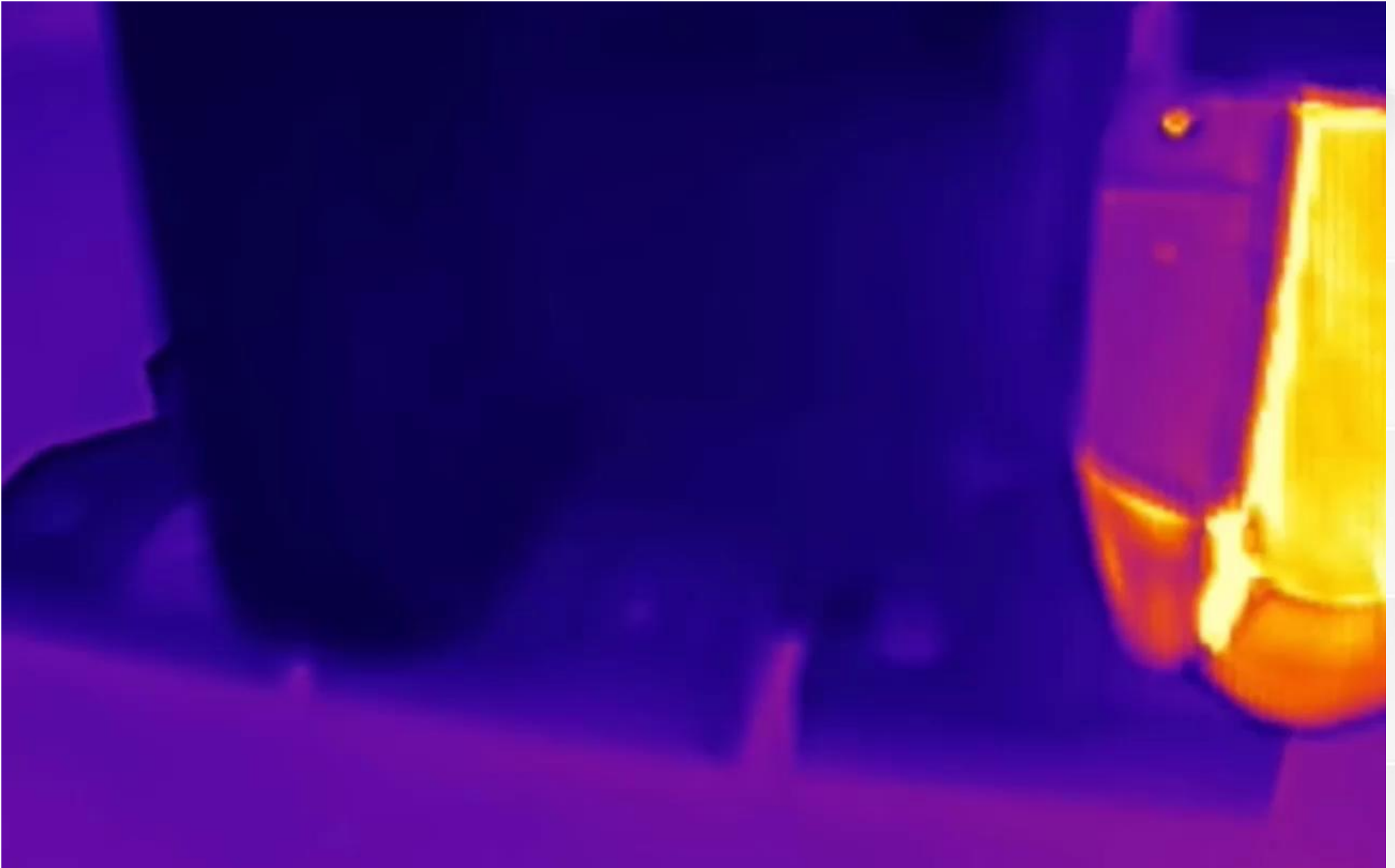
Slicer Layers

- Single layer preview.
- G-code shows what the printer is actually doing (lower left).
- *G1* = Move. Include X, Y, or Z coordinates.
- *G1 X Y E* = Move and extrude.

(Animation,
FlyingGyroscope GitHub)



(Animation,
Thermal camera
video by
NeedItMakelt)



Easy Filaments

- **PLA** (derived from corn starch).
 - General-purpose filament for indoor applications.
 - Excellent bridging.
 - Very stiff.
 - Creeps. Deforms permanently under load.
 - Low temperature resistance. Softens inside a car on a hot summer day.
- **PETG** (plastic water bottles).
 - Pretty good general-purpose filament.
 - Hygroscopic. Needs low-humidity storage.
 - Elastic. Excellent choice for springs.
 - Moderate temperature resistance. Can be used outdoors.

Easy Filaments

- What makes PLA and PETG easy?
 - Low warp. Low shrink.
 - High dimensional accuracy.
 - Room temperature print chamber. Open-air, non-enclosed.

Proper Protocol

Let's review some everyday guidelines.

- Secure the loose end of a spool of filament at all times.
 - Tangles ruin prints and waste time.
- Do not reach into the printer while it is running.
 - Use touchscreen or buttons to pause and resume prints.
- Watch the first layer before you walk away.
- Avoid burns.
 - Read nozzle and print bed temperatures on the display.
 - Never reach near a hot nozzle (210 °C = 410 °F).
- Do not leave fingerprints on print sheets.

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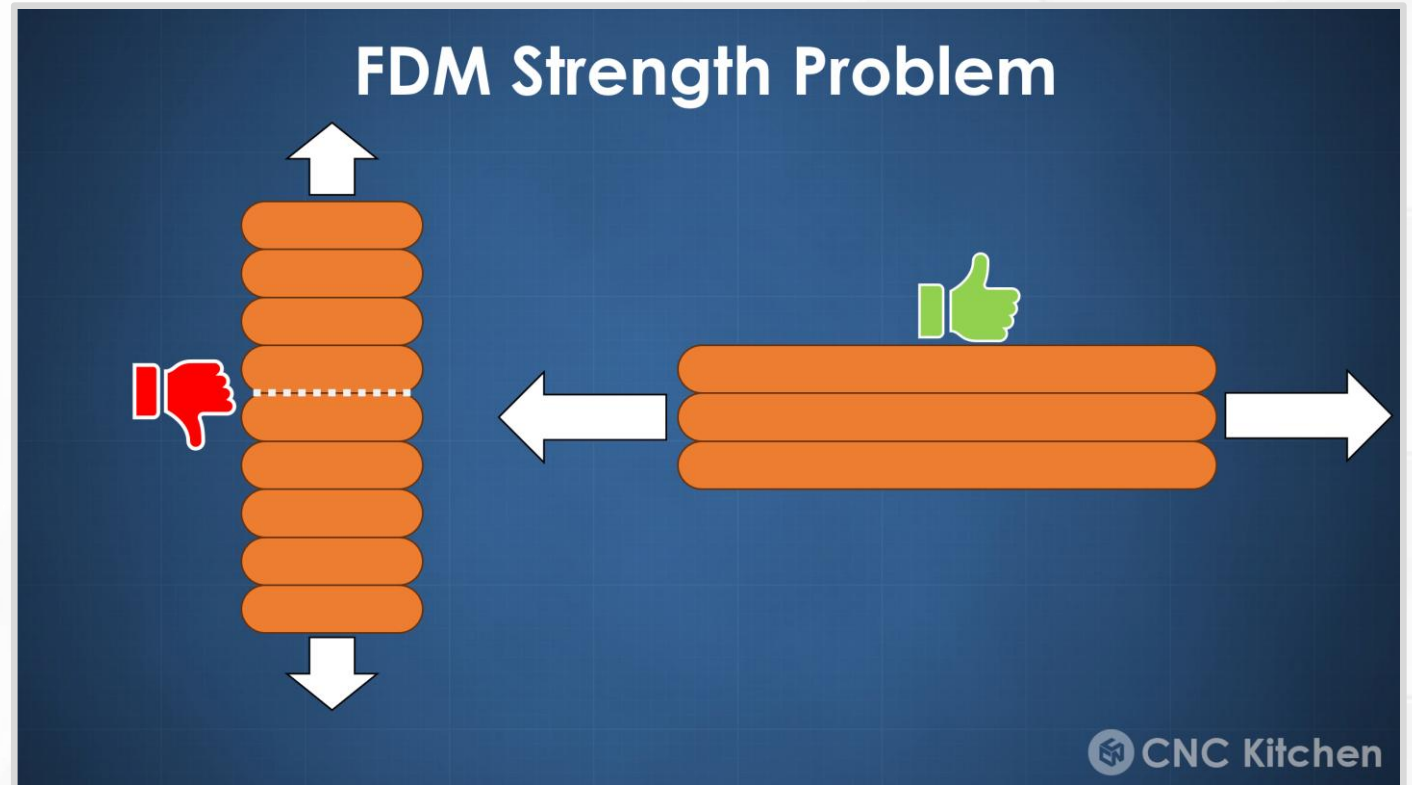
Thermal Expansion

Thermal Cycle

- Some plastics shrink as they cool.
 - Up to 1 or 2%.
 - Seems small, but large enough to distort dimensions.
 - Leads to cracking and delamination in extreme cases.
- Advice?
 - Tightly control ambient temperature and cooling rate.
 - Some filaments (ABS, Nylon) like it hot → Enclosed print chamber, extra heating.
 - Some filaments (PLA) like it cold → Open-air printing, extra cooling.
 - PETG is somewhere in the middle.

Anisotropic Strength

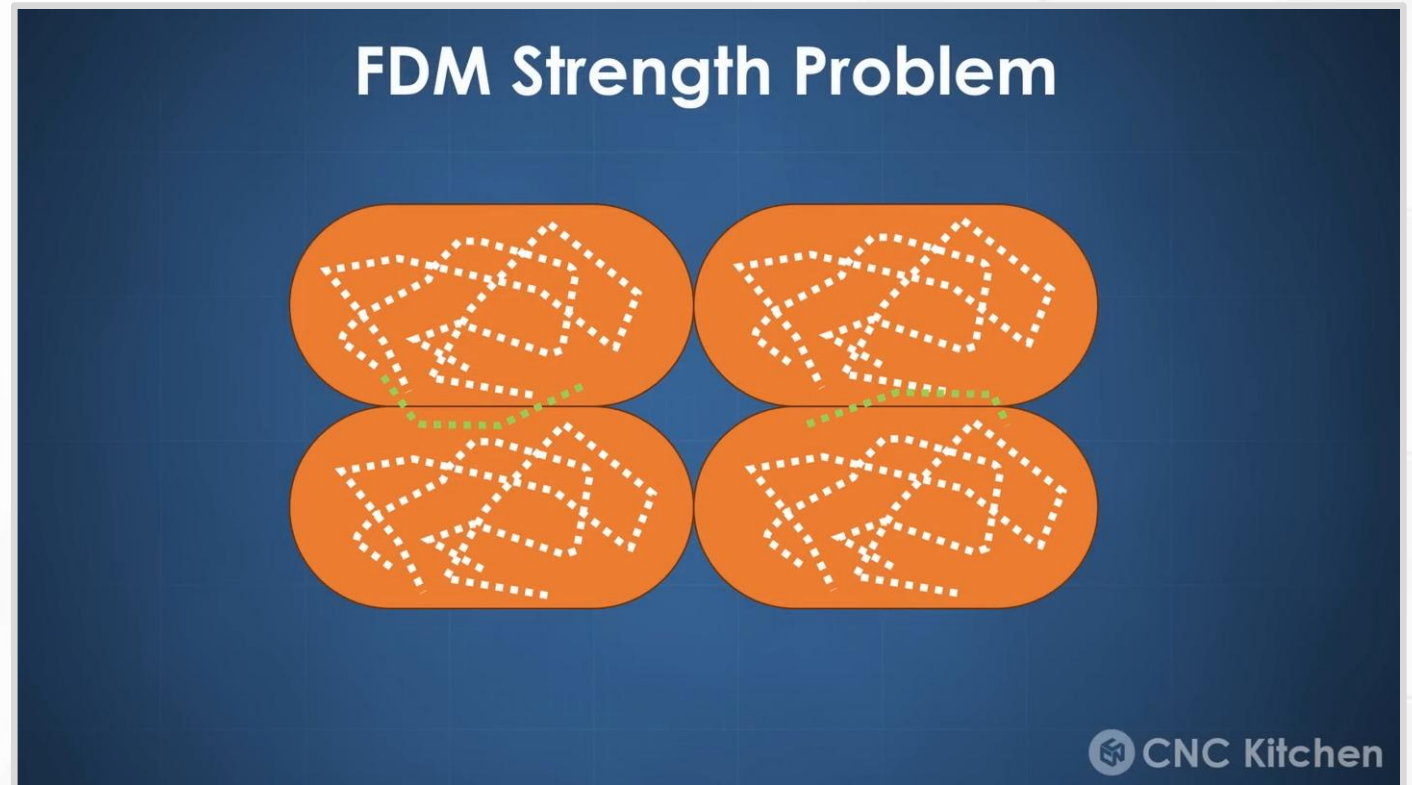
- Print orientation and layer lines affect strength.
- Layer-to-layer bonding is a weak point!
- And those weak points align.



(YouTube video by CNC Kitchen)

Anisotropic Strength

- Within an extrusion, hot filament bonds to hot filament.
- For a new layer, hot filament partially bonds to a cold layer.
- The “wrong” orientation can cost around 50% strength.



(YouTube video by CNC Kitchen)

Anisotropic Strength

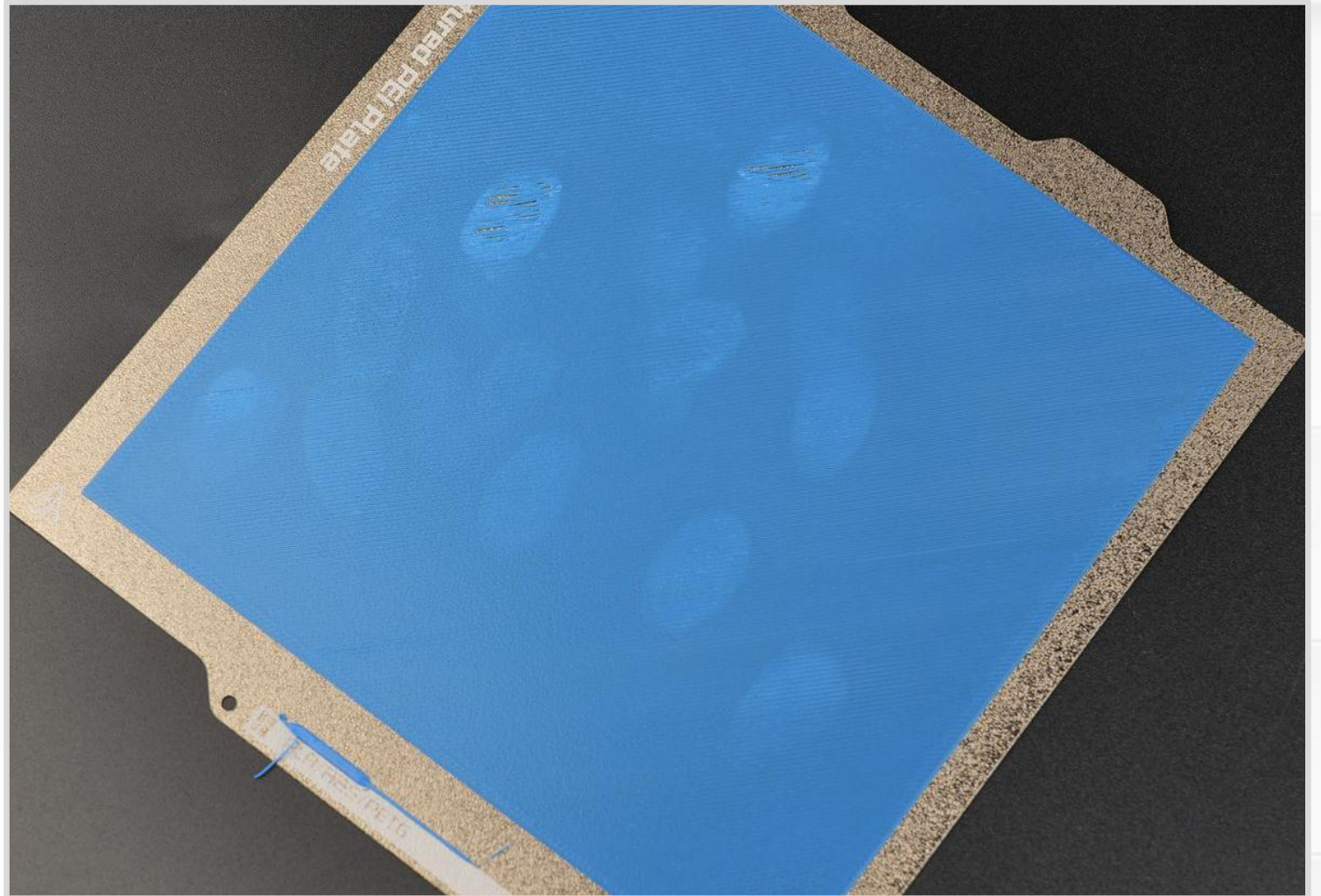
- Advice?
- Rotate prints to align layer lines and forces.
- Avoid tall and skinny shapes.
- Improve layer bonding.
 - Increase printing temps.
 - Decrease printing speed.

First Layer

- What is the most important layer in a 3D print?
 - The first layer!
 - First layers keep everything locked in place.
- Advice?
 - Always watch the first layer.
 - Tune first layer squish (extremely sensitive).
 - Flat surface area on the bottom of the 3D object.
 - Heated print bed.
 - Perfectly clean print bed.
 - Apply adhesion promoters (glue).

First Layer Mistakes

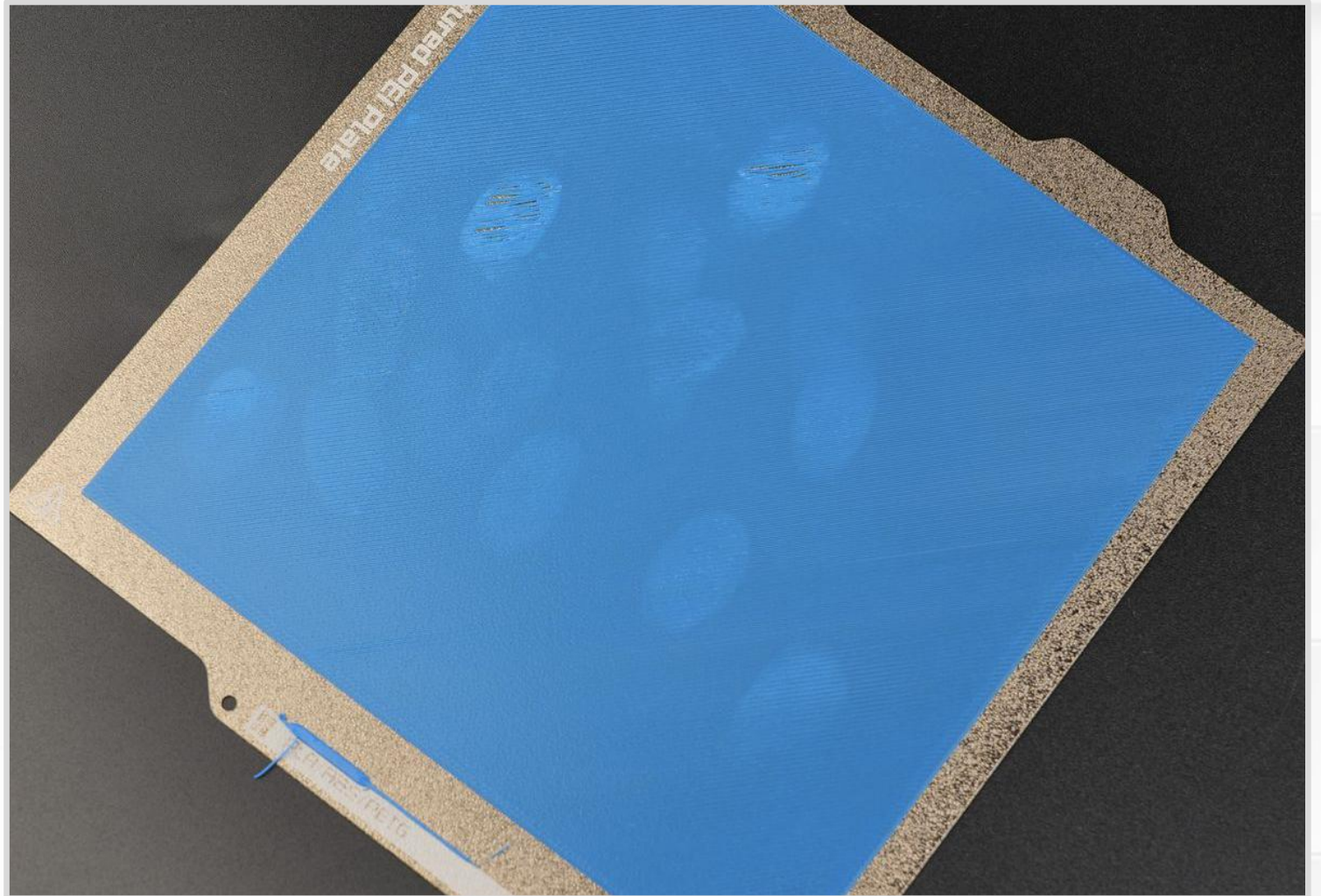
- What is wrong with this picture?
- **Fingerprints!** Plastic does not stick to oily fingerprints.



(Picture from [Bambu Labs Wiki](#))

First Layer Mistakes

- Advice?
- Deep cleaning: dish soap, soft brush, and water at the sink.
- Everyday use: window cleaner with ammonia.



(Picture from [Bambu Labs Wiki](#))

First Layer Mistakes

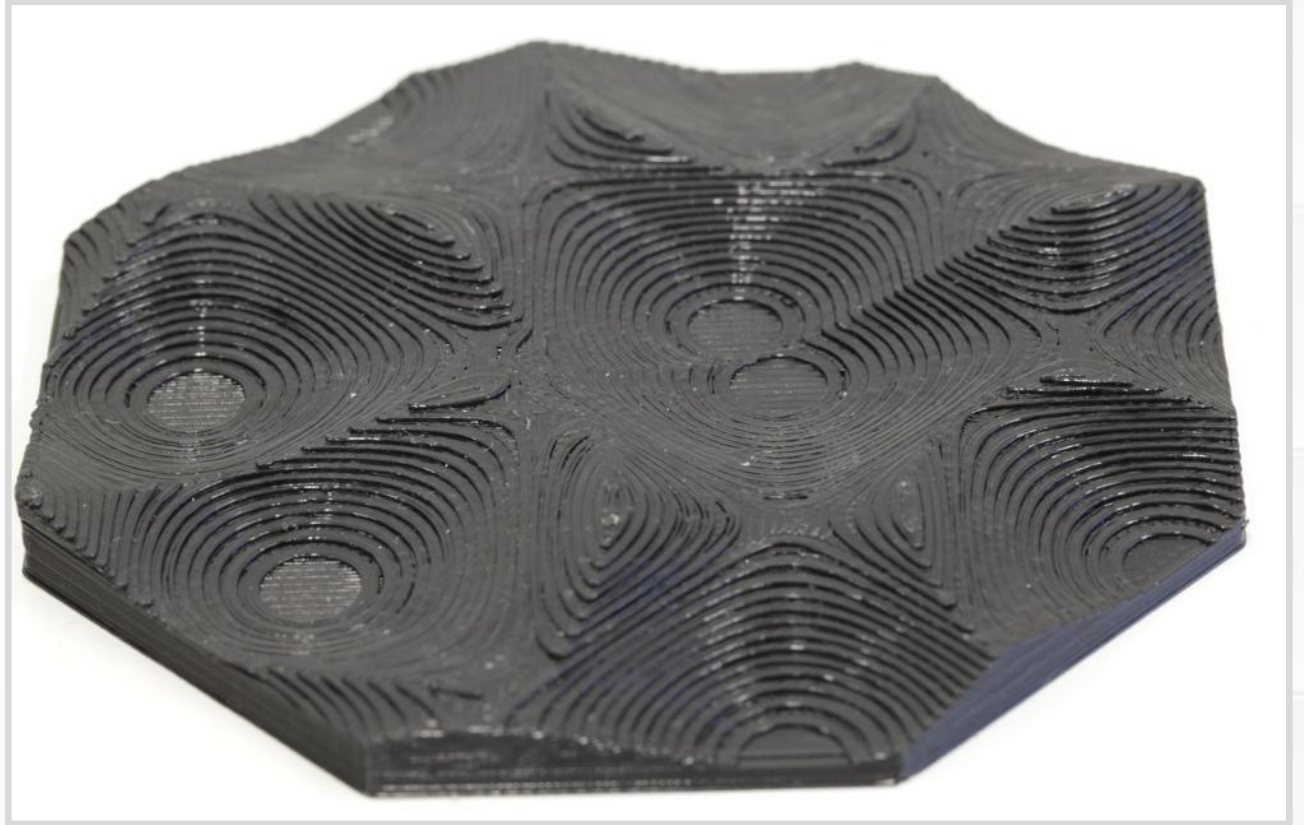
- What happens when a first layer does not stick?
- Print breaks free and makes “spaghetti”.
- When did things go wrong?
- Look at the partially completed, rectangular print.



(Screenshot from Everything STEM)

Stair-Stepping

- Printing with layers → Vertical slopes turn into stair steps.
- Mostly cosmetic effect.
- More pronounced on shallow angles.
- Advice?
- Rotate objects to hide layer lines.
- Smaller layer height at the cost of increased print time.



(Picture from Hamburg University Research Project)

First Layer Warping

- Why do corners and edges peel up?
- Regions shrink at different rates → Internal stresses → Stresses concentrate in corners.
- Shrinking is a percent → Large parts have the most problems.



(Picture from [Creality Cloud Blog](#))

First Layer Warping

Consider this analogy.

- Imagine you want to remove a label or piece of tape.
- What do you do?
- Start peeling up the corner because that is the weak point.
- Same thing here.
- Physics found the weak point.



(Picture from [Creality Cloud Blog](#))

First Layer Warping

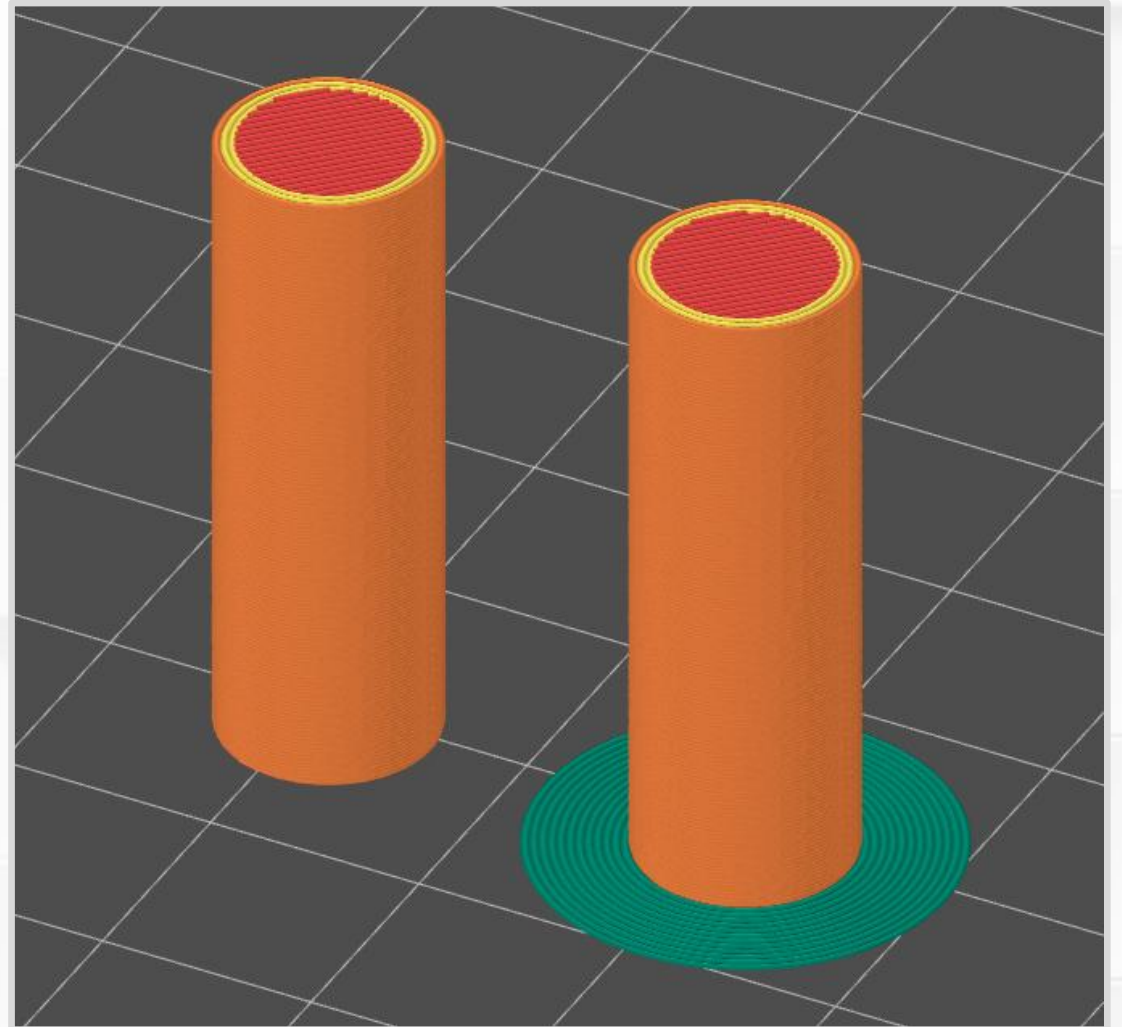
- Advice?
- Better control over temperatures.
 - Filament temperature.
 - Print chamber temperature.
 - Cooling fans.
- Evenly distributed cooling.
 - Avoid fans or drafts that cool one side.
- Round bottom corners to reduce stress concentration.



(Picture from [Crealitty Cloud Blog](#))

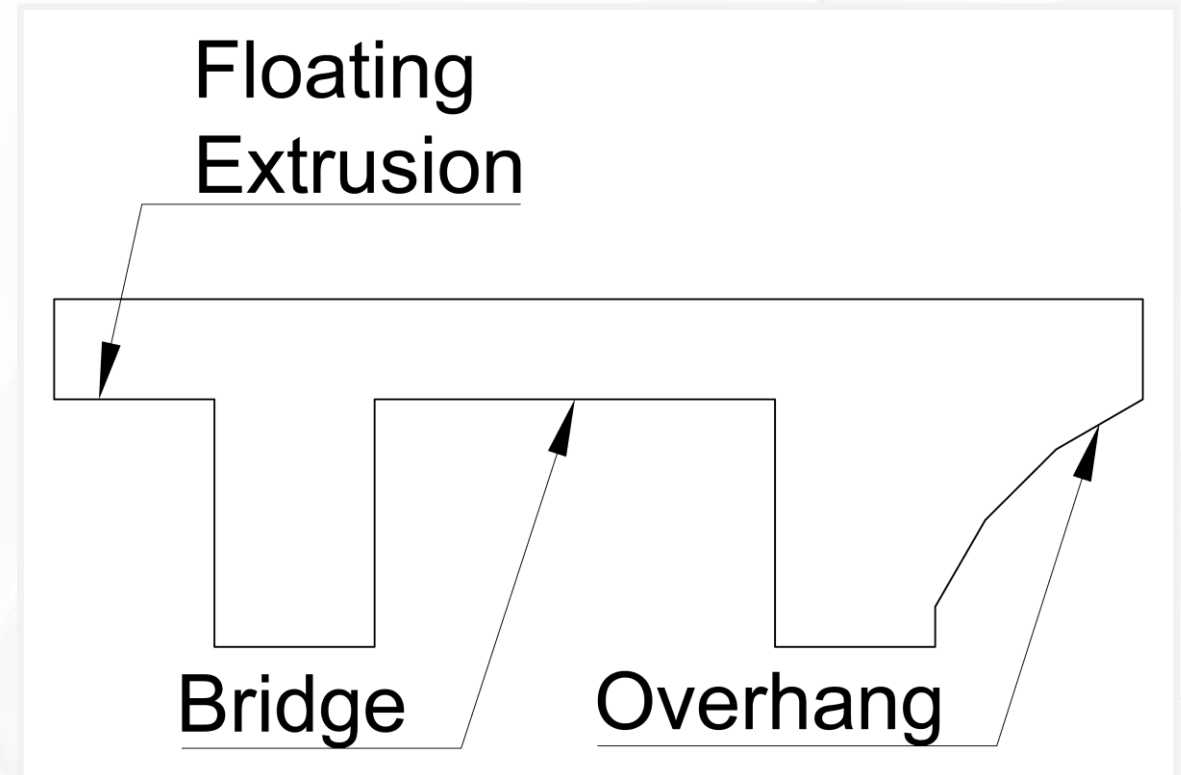
Brim

- What if you could add extra surface area to fight warping?
- Brim, like the brim on a hat.
- Concentric extrusions joined to bottom perimeter. (Green lines.)
- Added with a slicer.
- One layer tall.
- Easily peeled away after printing.



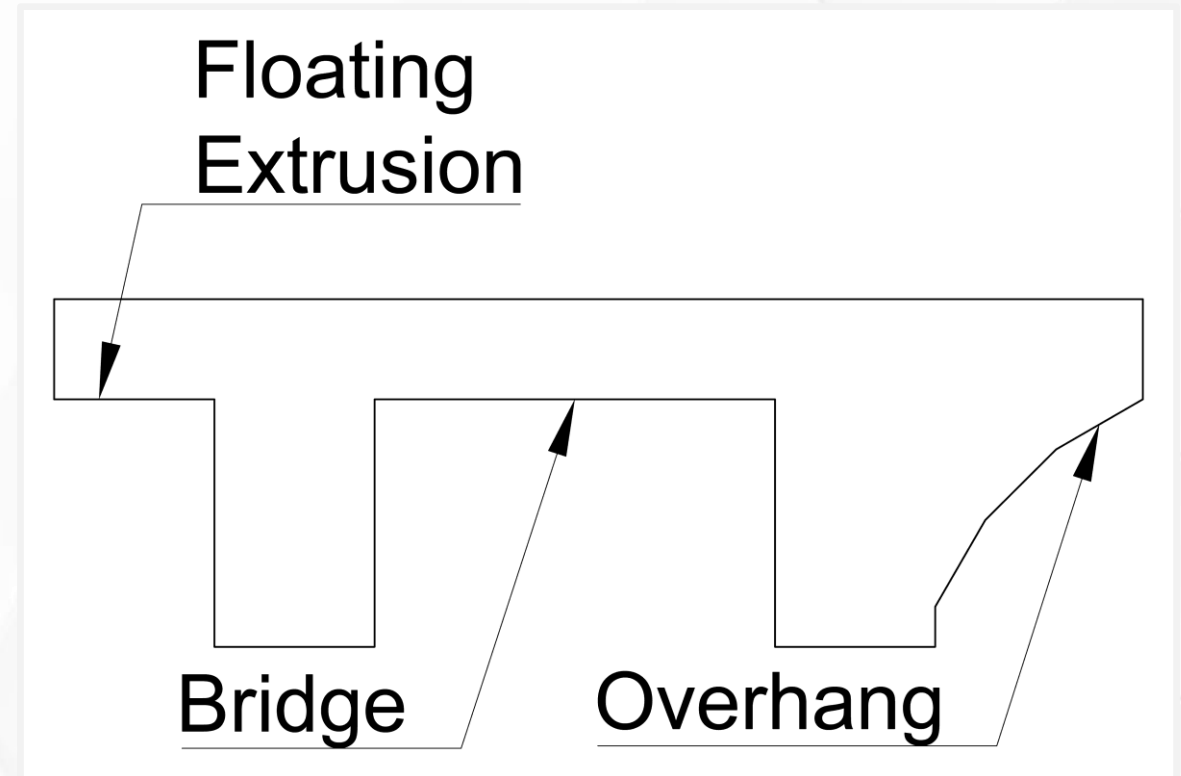
Levitating Extrusions

- *Critical thinking:* What happens when you try to levitate extrusions?
- Extrusions work best when you press them onto a solid surface.
- Predict what this print will look like.
- What parts will be the most deformed?



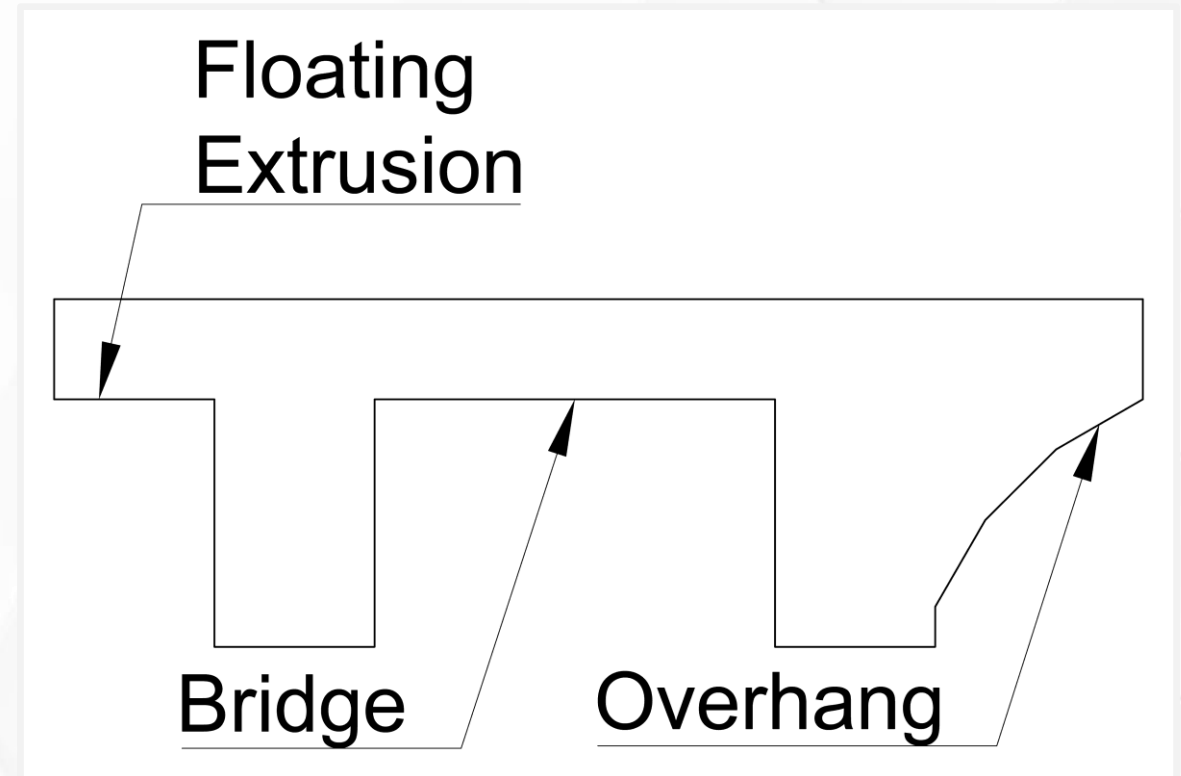
Overhang

- Extrusions that lean over an edge.
- Only the outermost extrusion is affected.
- Partially supported along one side.
- Advice?
- Keep overhangs at 45° to be safe.



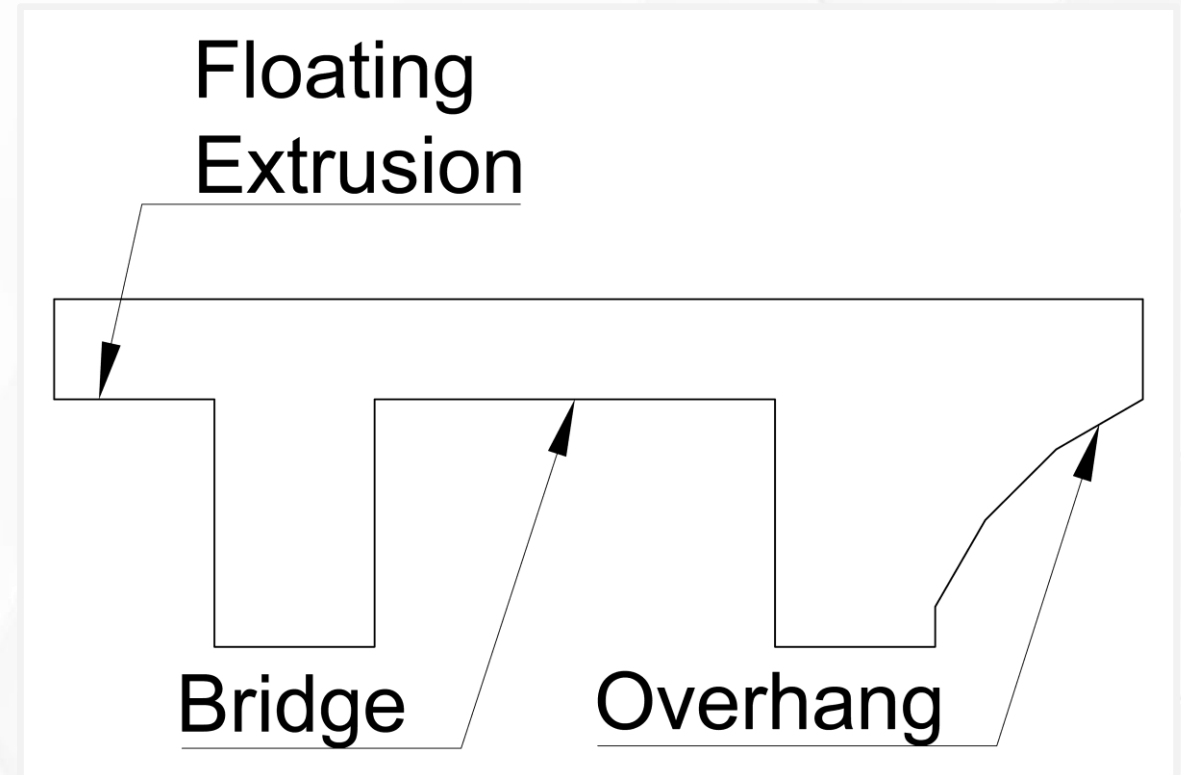
Bridge

- Both ends of an extrusion are supported and anchored.
- Middle hovers over a gap.
- First few layers sag and droop down.
- Eventually recovers to flat layers.
- Advice?
- Keep bridges as short as possible.



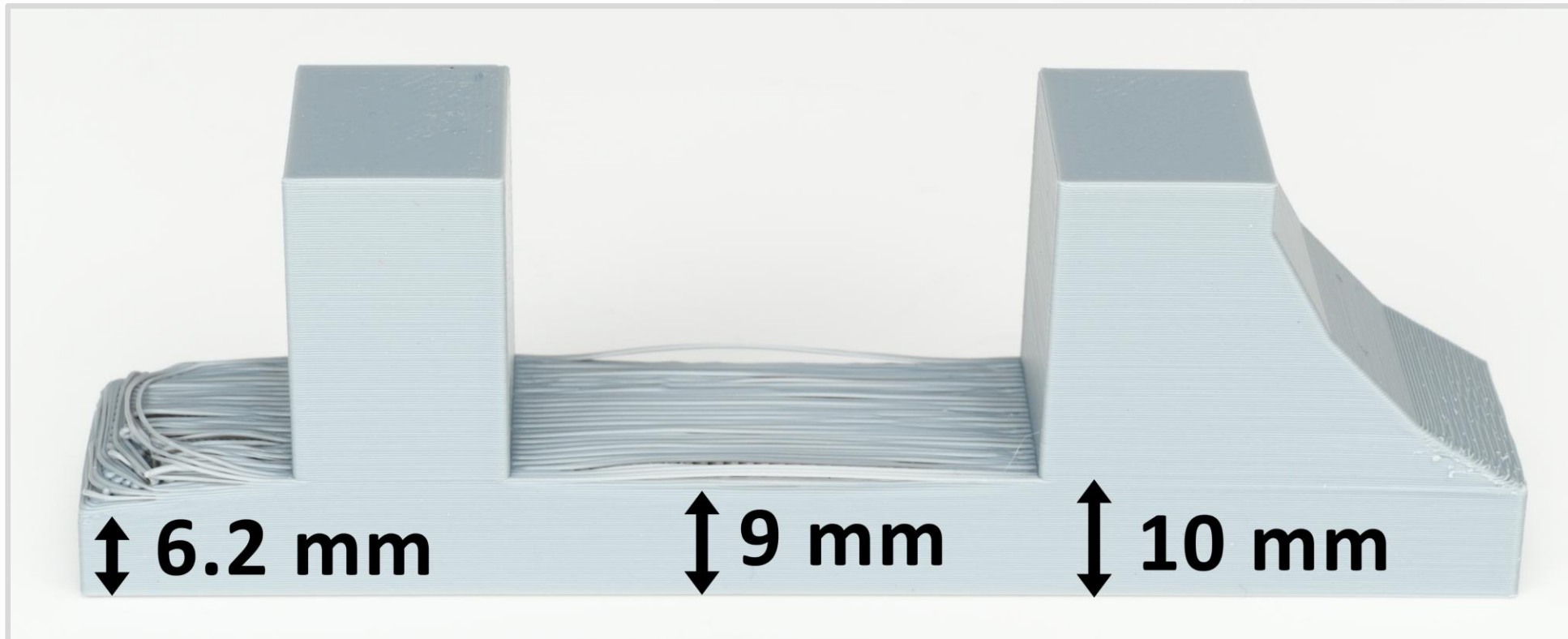
Floating Extrusion

- Extrusion that hangs in the air.
- At least one end is not anchored to a solid surface.
- Advice?
- **Avoid. Very unreliable.**



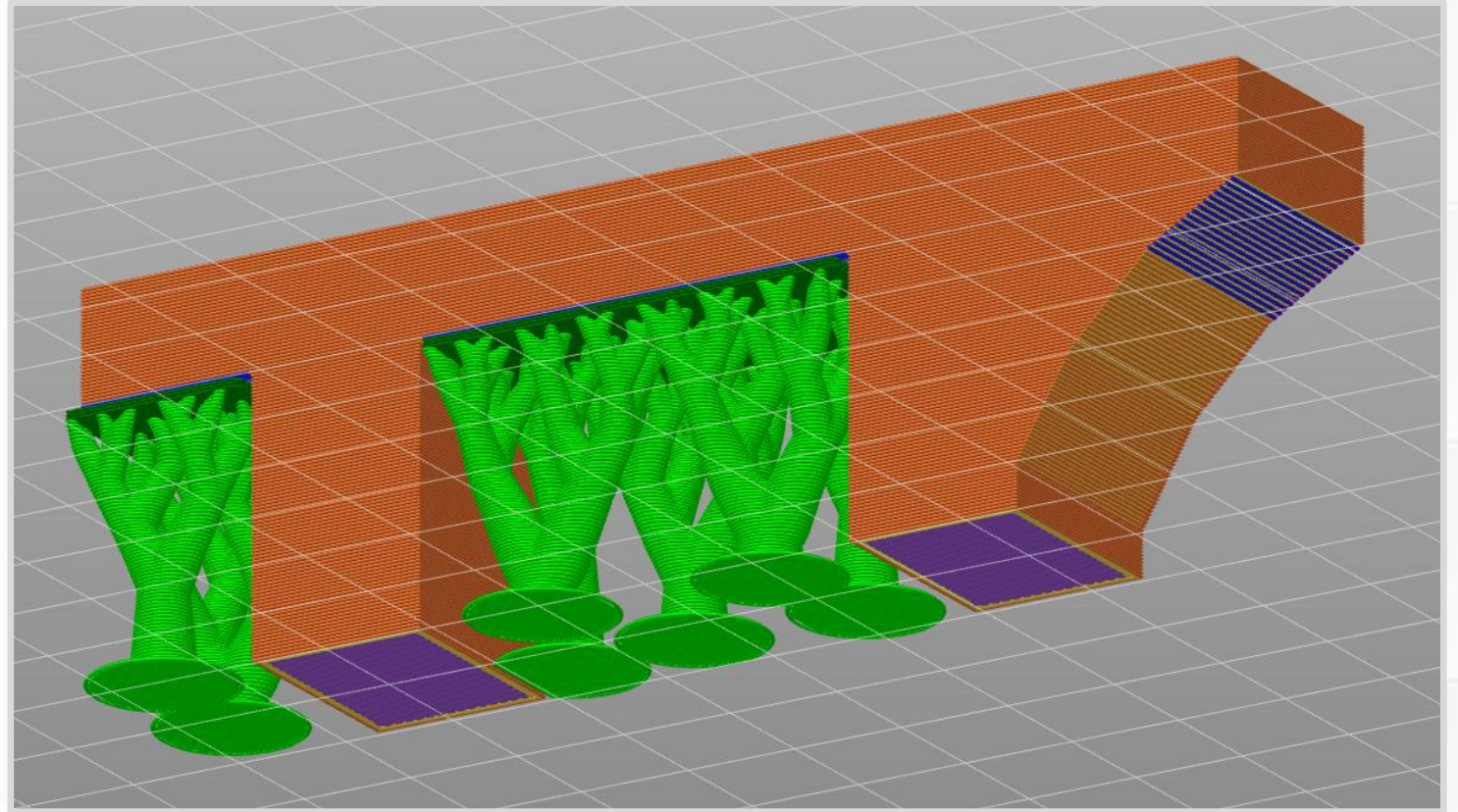
Stress Test Results

- All overhang angles look good.
- Bridge recovers to flat layers in 1 mm (**5 layers**).
- Floating extrusions recover in 3.8 mm (**19 layers**).



Supports

- Temporary scaffold that fixes:
 - Floating extrusions.
 - Long bridges.
 - Steep overhangs.
- Added by the slicer.
(Green trees-like branches.)
- Extra complexity, plastic, and print time.

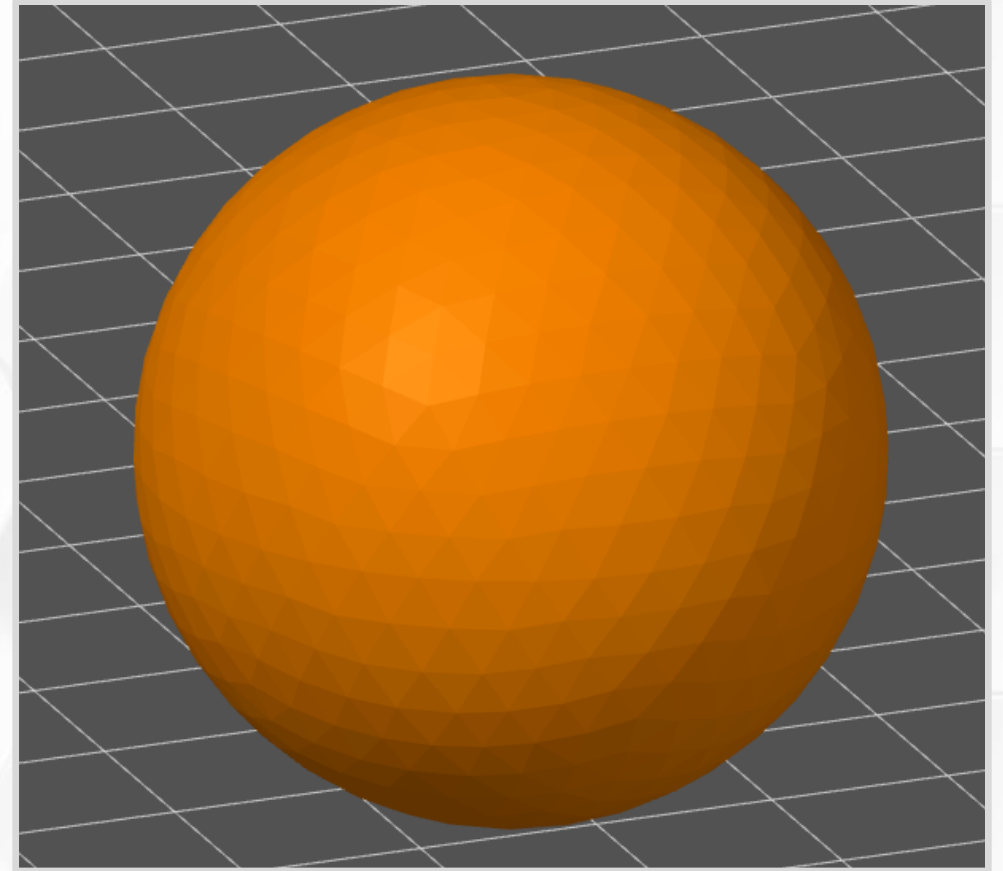


Supports

- In special situations, you can include supports as part of a 3D model.
 - *Example:* 3D Printing Trick by Clough42, and a follow-up video.
- Tree/organic supports can reach into difficult spaces.
- Supports are a delicate balance. Why?
 - If they do not bond well enough, they hold nothing in place.
 - If they bond too well, they become part of the print.
- Supports can leave a rough surface finish.
- Sometimes you need to cut away pieces of support.

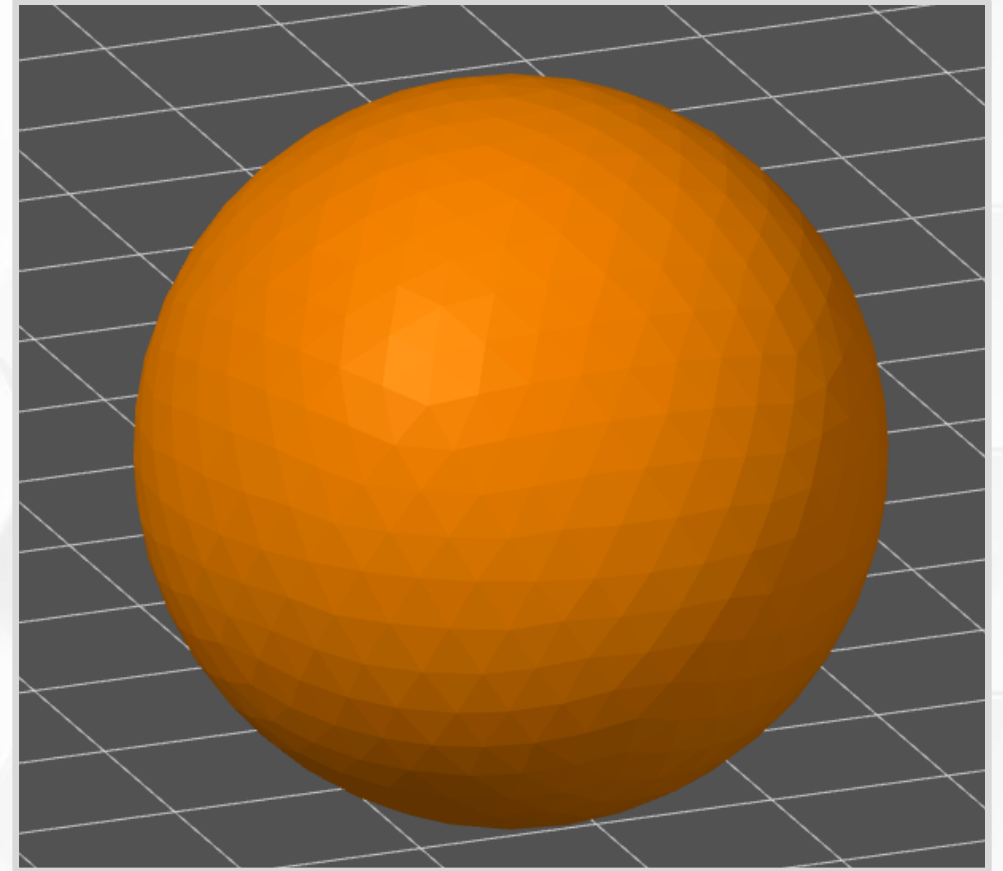
Limitations Review

- **Printability** summarizes limitations for a specific object.
- Without extra considerations, a simple sphere has poor printability.
- What limitations do you see?



Limitations Review

- Very bad first layer contact. Significant spaghetti risk.
- Steep overhangs on the bottom.
- Second layer is steep enough to create floating extrusions.
- Stair-stepping creates uneven surface finish.
- How can we fix this print?



Limitations Review

- Brim is not helpful.
 - Object starts with too little surface area.
- Add supports to the bottom.
- Cannot fix stair-stepping by changing print orientation.
 - Use small layer heights to reduce visual artifacts.
 - Use adaptive layer heights to make the surface look more uniform.
- Print two halves, then put them together.
 - Add tabs to help you align them.

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Dimensional Accuracy

- Be careful modeling parts with exact dimensions.
- Why?
 - 3D prints are imperfect.
 - When assembled together, the fit is probably too tight.
- Where do these errors come from?
- And how do we fix it?



Dimensional Accuracy

- Sources of error include but are not limited to:
 - Parts that shrink or warp.
 - Inconsistent extrusions.
 - Racked gantry (skewed axes).
 - Shipping damage.
 - Incorrect belt tension.
 - Backlash, slop, or loose parts.
 - Moving parts that are worn down.
 - Sloppy slicer settings.
 - Thermal expansion (*discussed in Part 3*).
 - Vibrations (*discussed in Part 3*).

Modeling Calibration

- *(Printed size) = (Size) × (Scale factor) + (Constant offset).*
- Simplified model.
 - Assume fully functional motion system.
 - Apply to X and Y dimensions.
- Scale factor. Error changes with size.
 - Part shrinkage.
- Constant offset. Error is always the same.
 - Extrusion inconsistencies. Over and under-extruding.
 - Surface roughness. Additives like carbon fiber, wet filament, over-extruding.



(xkcd.com)

Fixing Scale Factor

- **Adjust part scaling in the slicer to fix scale factor.**
 - Scale XY size, or
 - Dedicated shrinkage compensation setting.
- Applies to
 - Large parts over 6 inches, 150 mm.
 - Filaments that shrink, like ABS.
- In my experience, safe to ignore for PLA, PETG, and small projects.

Fixing Constant Offset

- **Add clearance (air gap) between parts that touch to fix constant offset.**
- Very important for projects with assembled pieces!
- Best to add clearance while designing 3D objects.
 - Greatest flexibility. Independently adjust each surface.
- Or try adding negative XY offset in slicer.
 - Equally offsets all vertical surfaces.

Adding XY Clearance

- How much XY clearance? It depends ...
- Start with 0.25 mm. Offset each side by half, 0.125 mm.
- *Example:* Making a hole for a bolt.
 - Add 0.2 - 0.3 mm of clearance for a close fit.
- *Example:* Making a lid that easily slides into place.
 - Add 0.4 - 0.8 mm of clearance for a loose fit.

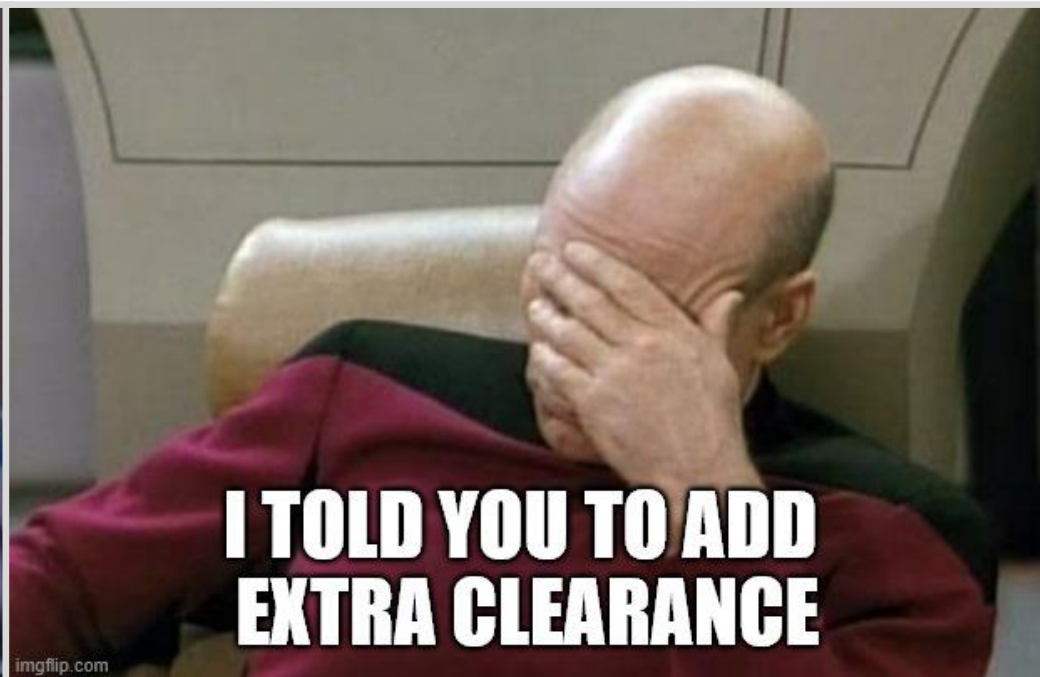
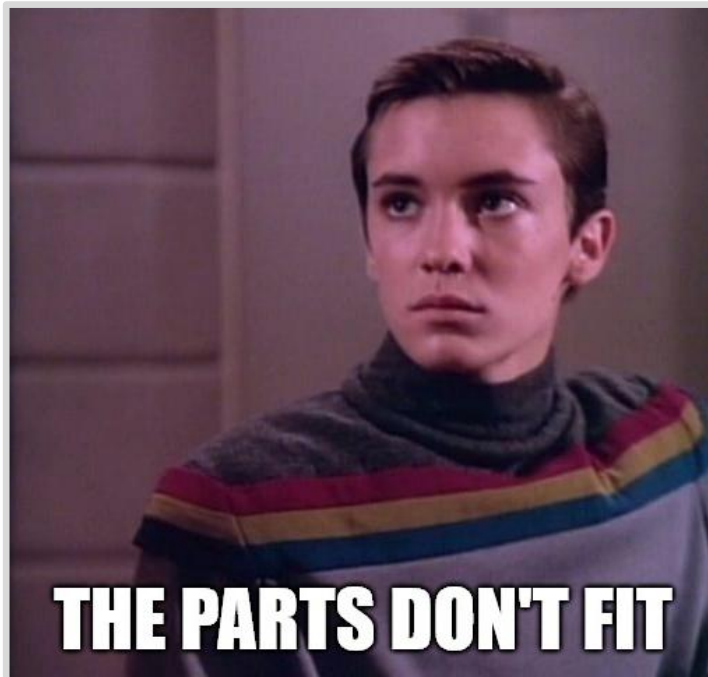
Adding Z Clearance

- How much Z clearance? It depends ...
- Start with one layer or 0.1 mm.
- Increase clearance in certain scenarios.
 - Sagging extrusions.
 - Rough bottom surface from supports.
 - High and low spots on print bed.
 - Tilted print bed.

Summary

If your project has assembled parts then ...

- You sometimes need scale factor.
- You always need constant offset.



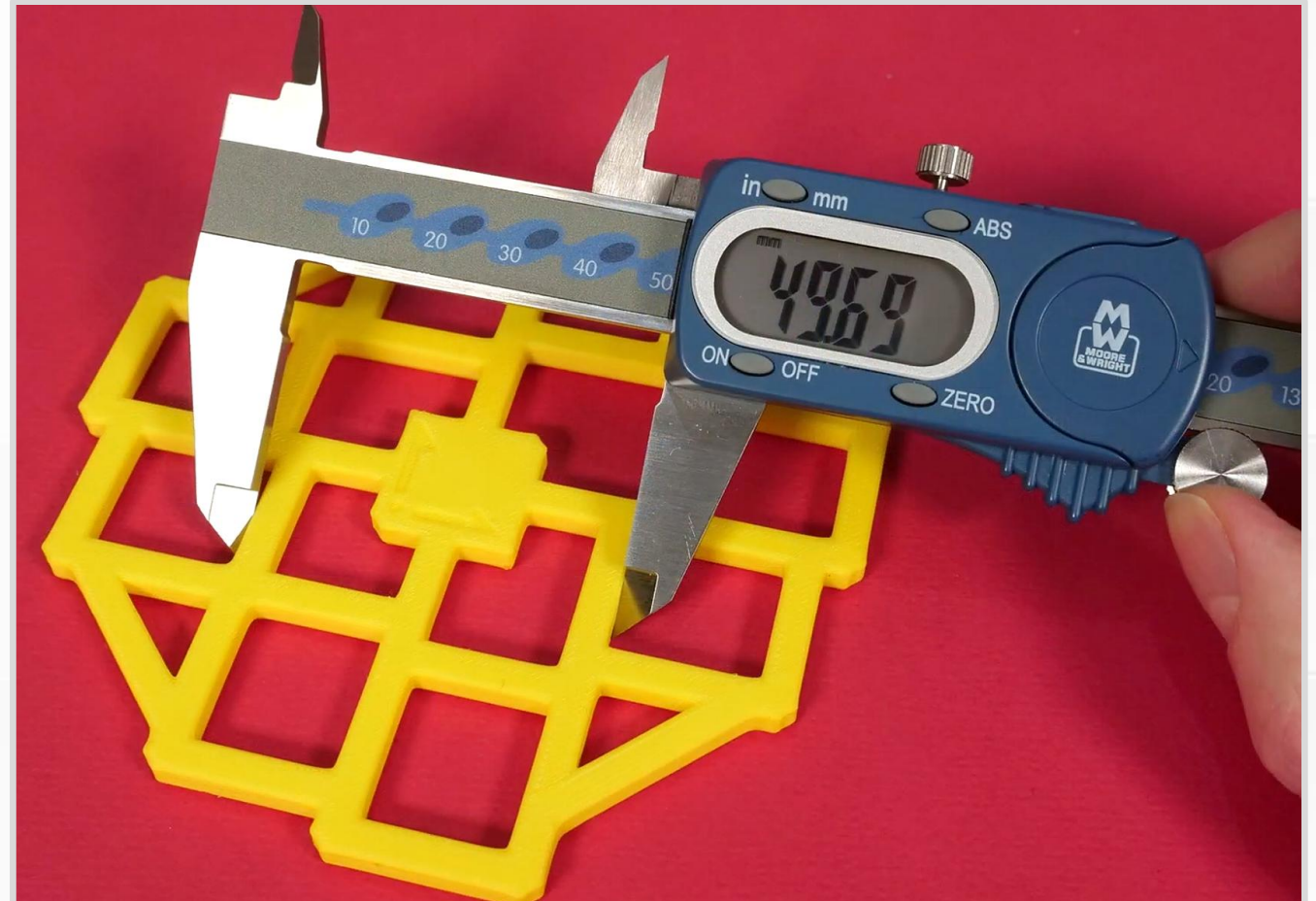
Calibration Prints

- Calibration cubes.
 - 20 mm cube.
 - Labels for X and Y-axis.
 - Tells you constant offset for outside dimensions.
- Clearance test by 3DMakerNoob.
 - Print-in-place clearance/tolerance.
 - Alters the gap between pieces.
 - Feel which pieces move freely.



Calibration Prints

- Califlower by Vector3D.
 - Reasonably priced. Includes walkthrough and calculator.
 - Inner dimensions.
 - Outer dimensions.
 - Shrinkage.
 - Skew, when rectangles turn into parallelograms.
- Calibration video by CNC Kitchen.



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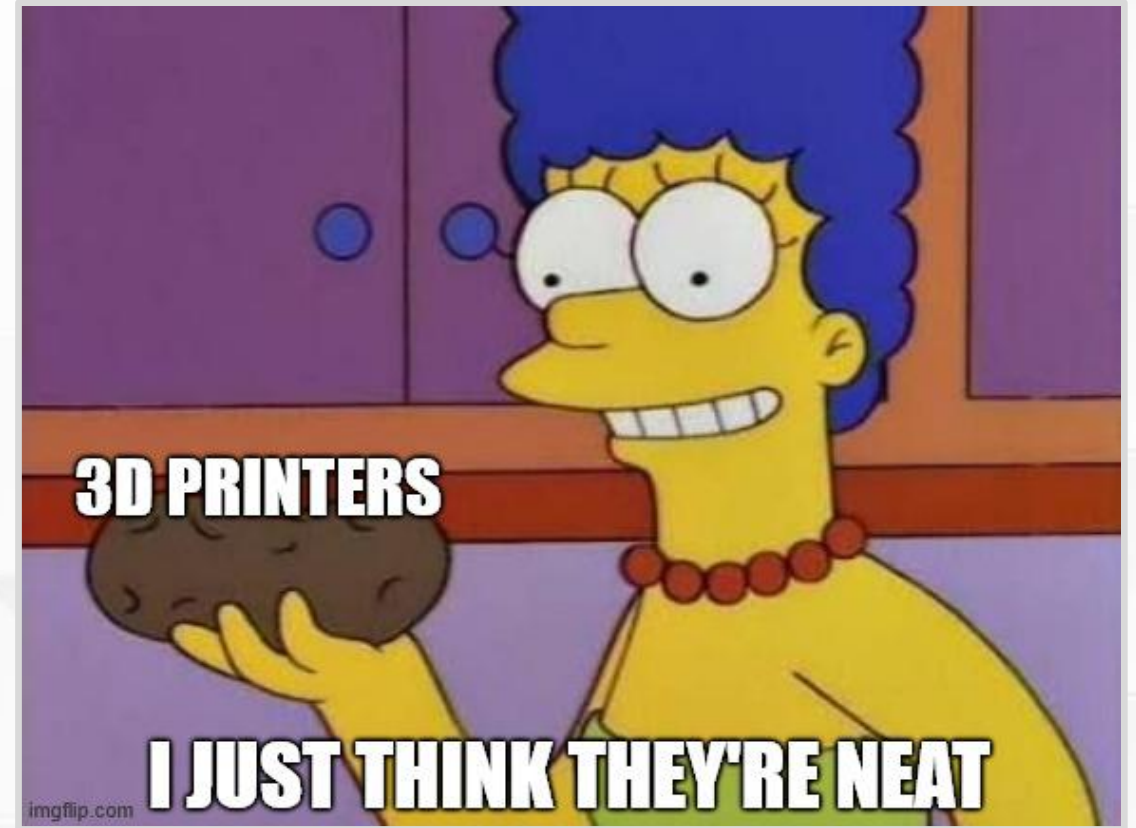
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An Amazing Tool

- Playground for **creative design** and **creative problem solving**.
- Massive utility for cheap.
- Make replacement parts to fix and repair.
- Bring your ideas to life.
- Fast turnaround time.
- Low barrier to entry.
- Practical for hobbyists, students, and professionals.

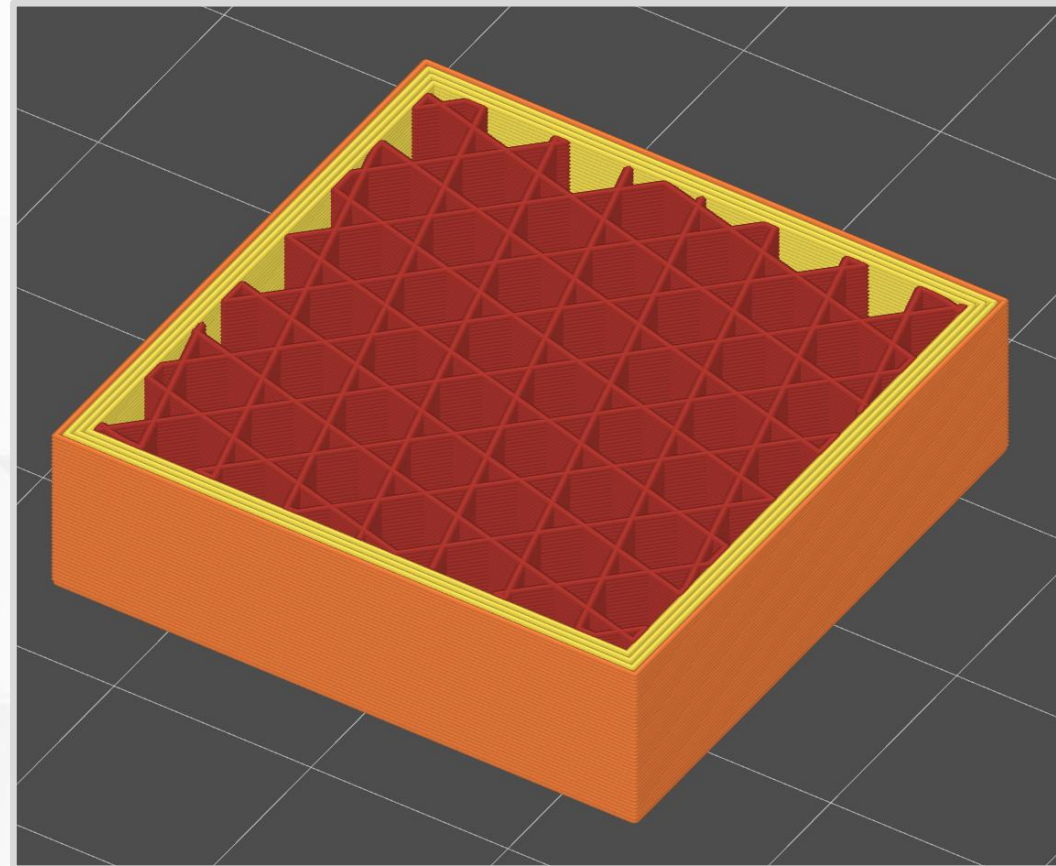


An Amazing Community

- RepRap festivals, maker fairs, local clubs and maker spaces, ...
- People share cool projects and ideas. Examples include:
 - Full spectrum printing*
 - BumpMesh.com*
 - Voron*
- YouTube makers with educational content. Examples include:
 - CNC Kitchen*
 - Maker's Muse*
 - JanTec Engineering*
 - 3D Printing Nerd*
 - Teaching Tech (retired)*

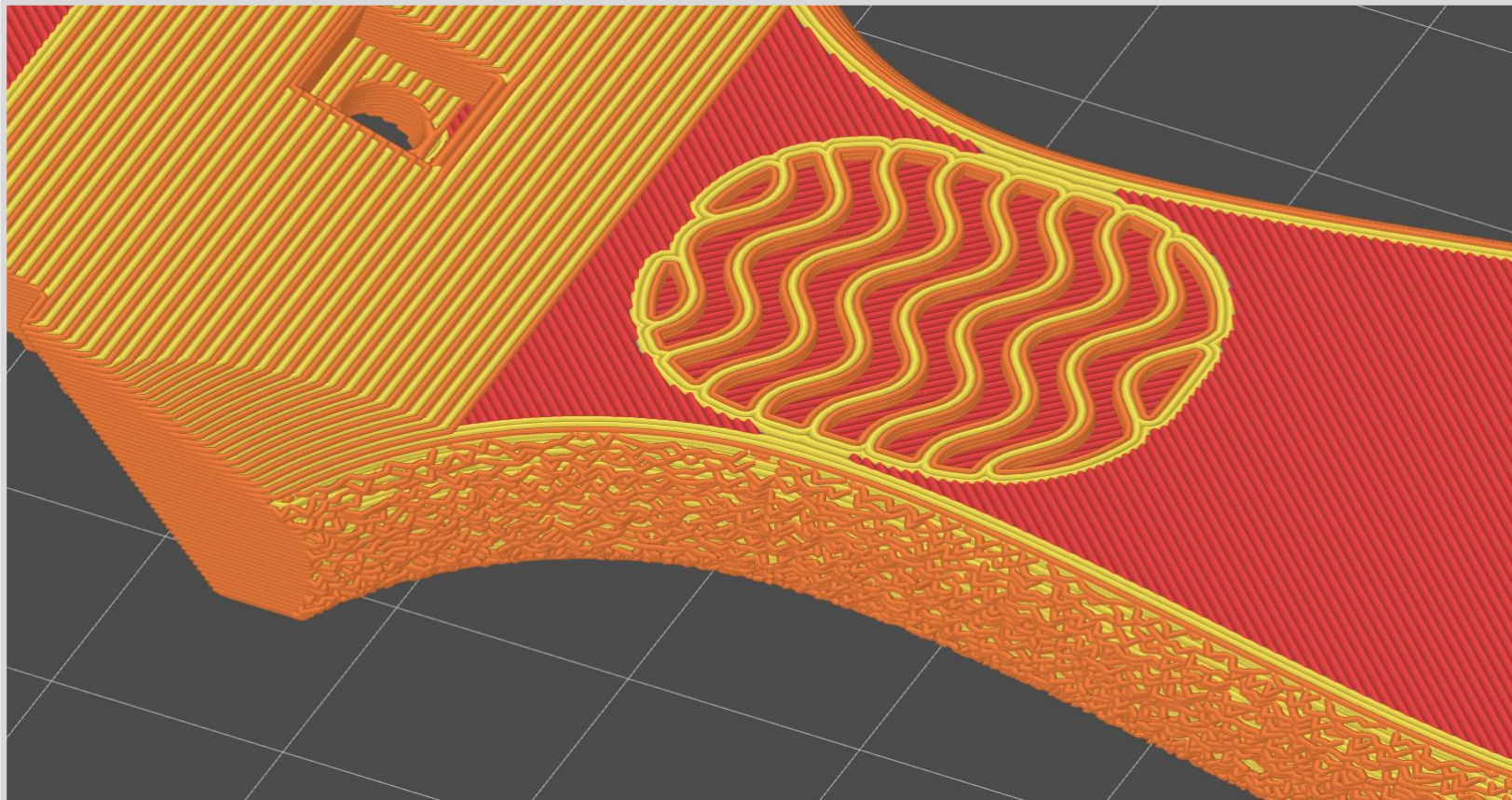
FFF Advantages

- Make lots of shapes with tiny “building blocks.”
 - Typical extrusions are
0.2 mm tall (1-2 pieces of paper)
0.4 mm wide.
- Not restricted to uniform, solid objects.
 - Easily create internal geometry.
 - Great at making hollow objects.



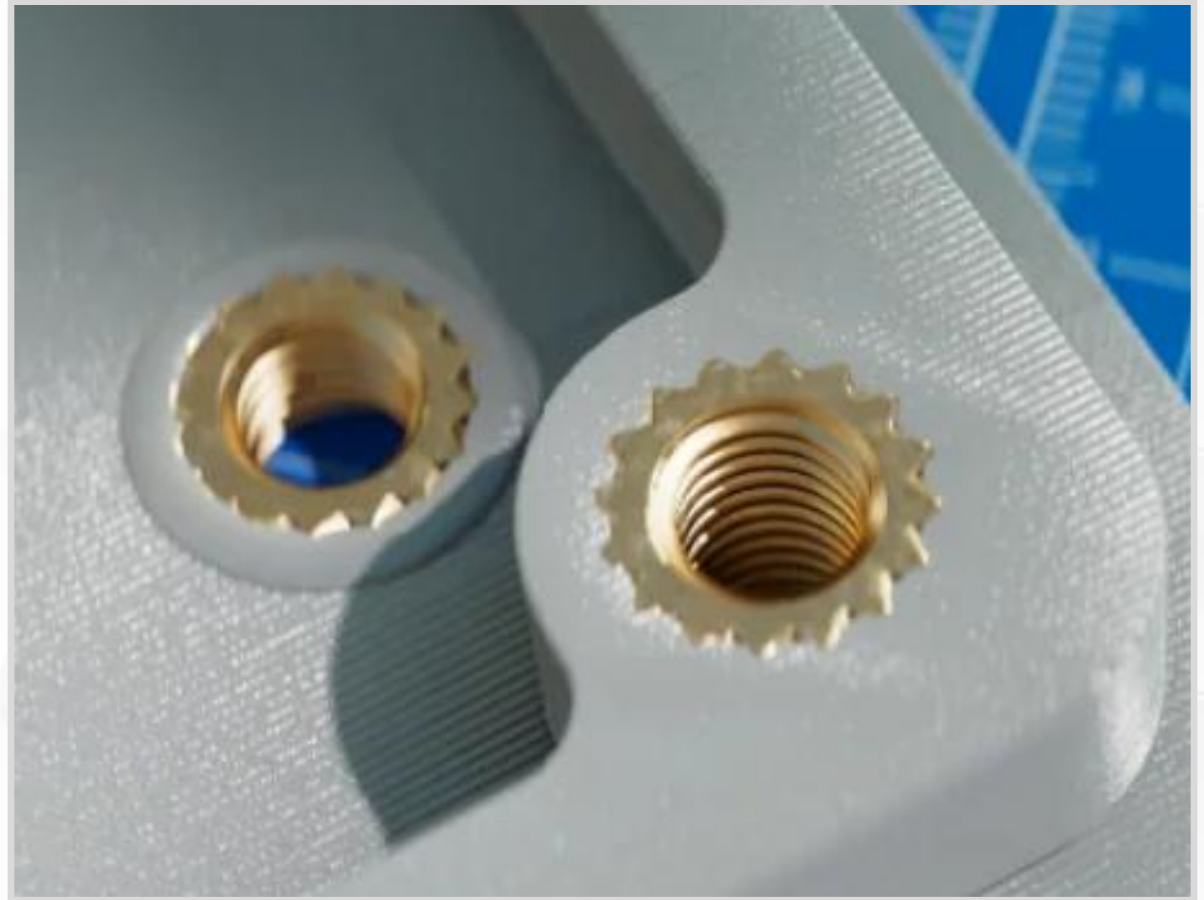
Custom Slicing

- Change slicer settings for specific regions or surfaces. Too many to list.
- *Example:* Add texture to handles with fuzzy skin (bumpy extrusions).



Heat Set Inserts

- Threaded anchor for screws. Why?
- Plastic is soft → Screws easily wear out plastic holes.
- Install with a hot soldering iron.
- Locks in place when it cools.
- Very durable.



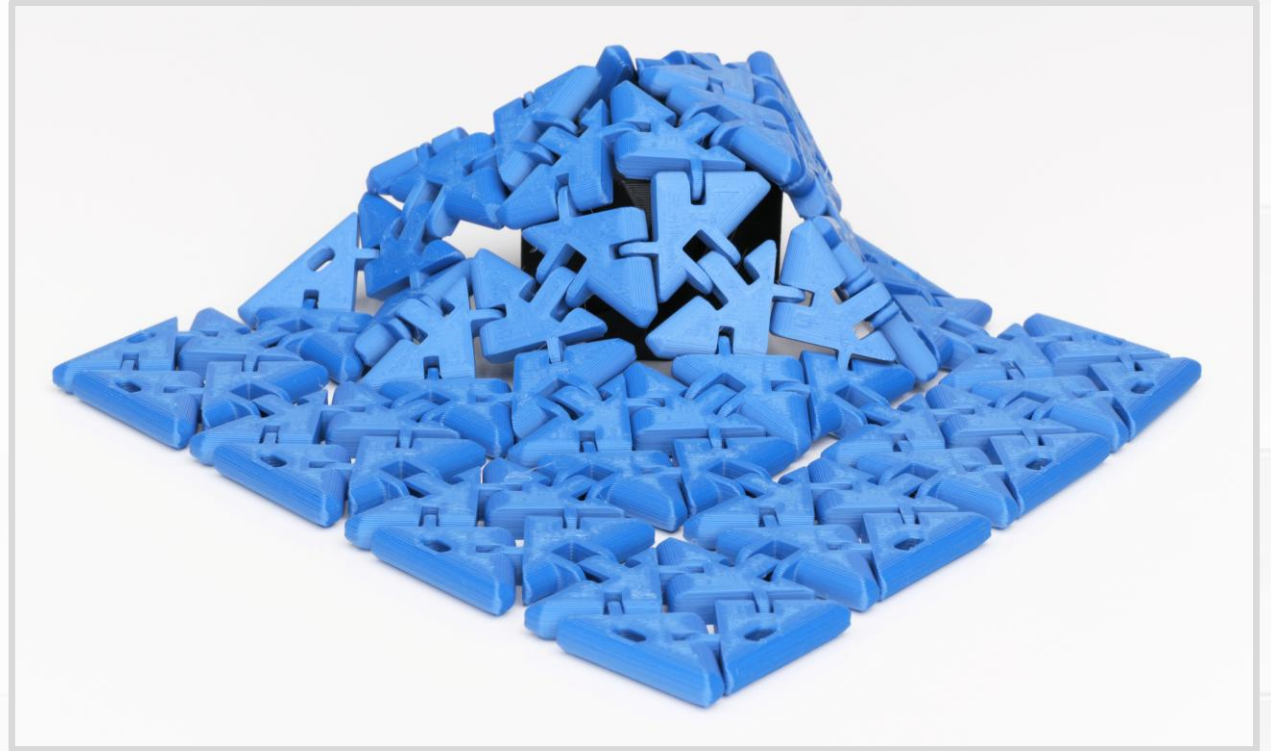
(Pictures from [CNC Kitchen Store](#))

Captive Parts

- Insert an object into a pocket or void in the middle of a print.
 - Nuts and fasteners.
 - Magnets.
 - Weights.
 - Reinforcing bar or plate.
- Locked inside when the print finishes.
- Object **must** sit below the current layer, or it creates a collision with the toolhead.
- A little inconvenient.
 - Need to be at the printer at the right layer.

Print-in-Place (PIP)

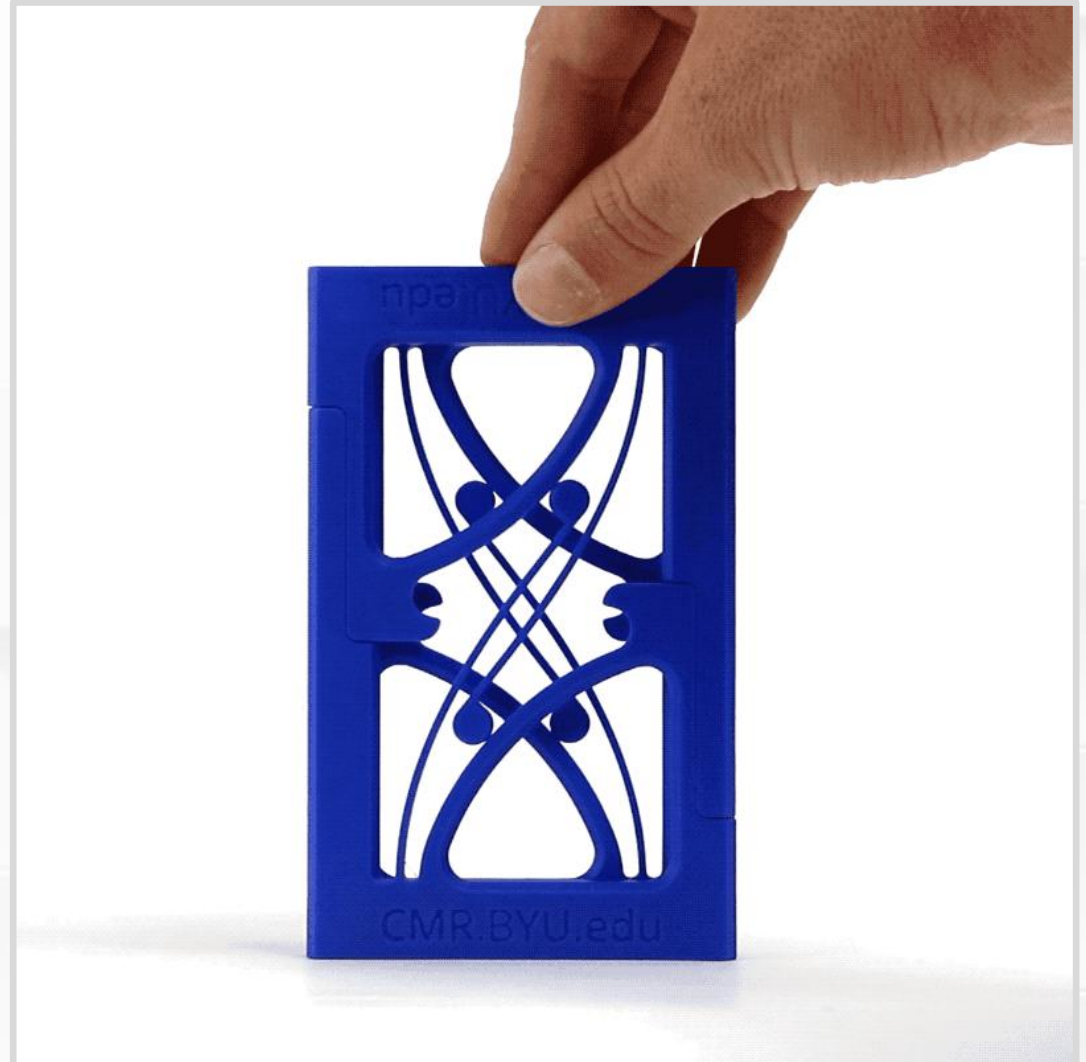
- Single print with interlocking parts.
- Assembled when the print finishes.
- *Example:* Hinges, chain links, chainmail.
- Flexify collection by [printables.com/@Flying Gyroscope](https://printables.com/@FlyingGyroscope)



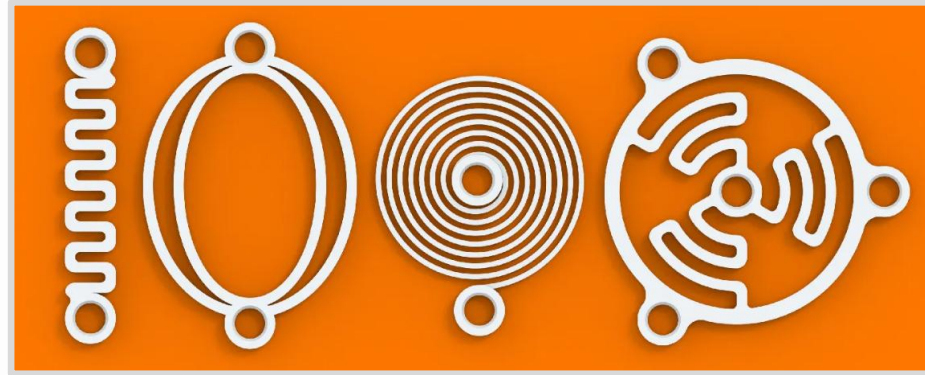
Functional Flexing

- Springs and **compliant mechanisms**.
- Features and shapes that function by bending and flexing.
- Elastic, springy materials.
 - No permanent deformation.
- Remember part strength and layer lines.

(Animation, Constant Force
Mechanism by BYU CMR)



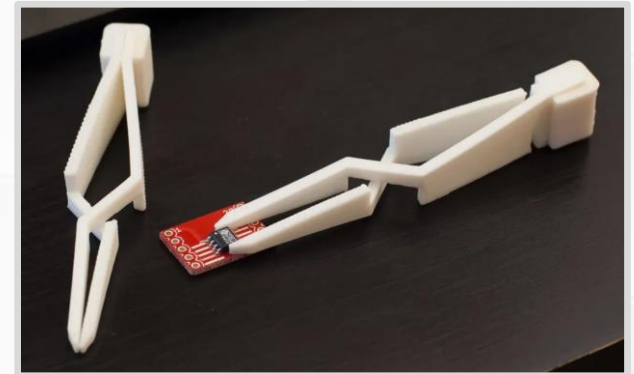
Springs by VC Design



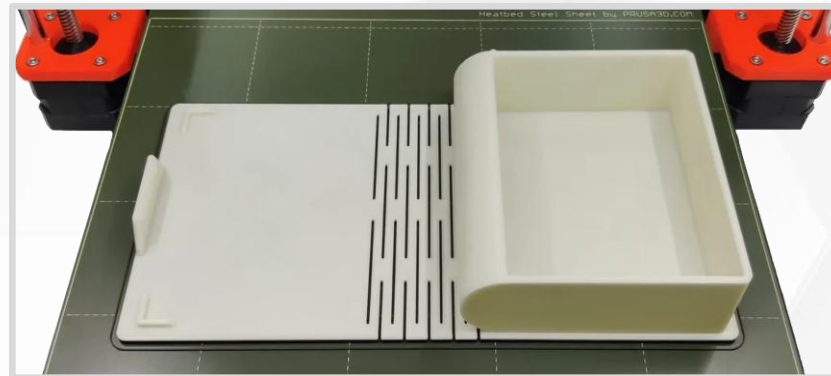
Clip by AE



Cross tweezer by joo



Living hinge by DrJones





github.com/FlyingGyroscope/IntroTo3DPrinting/